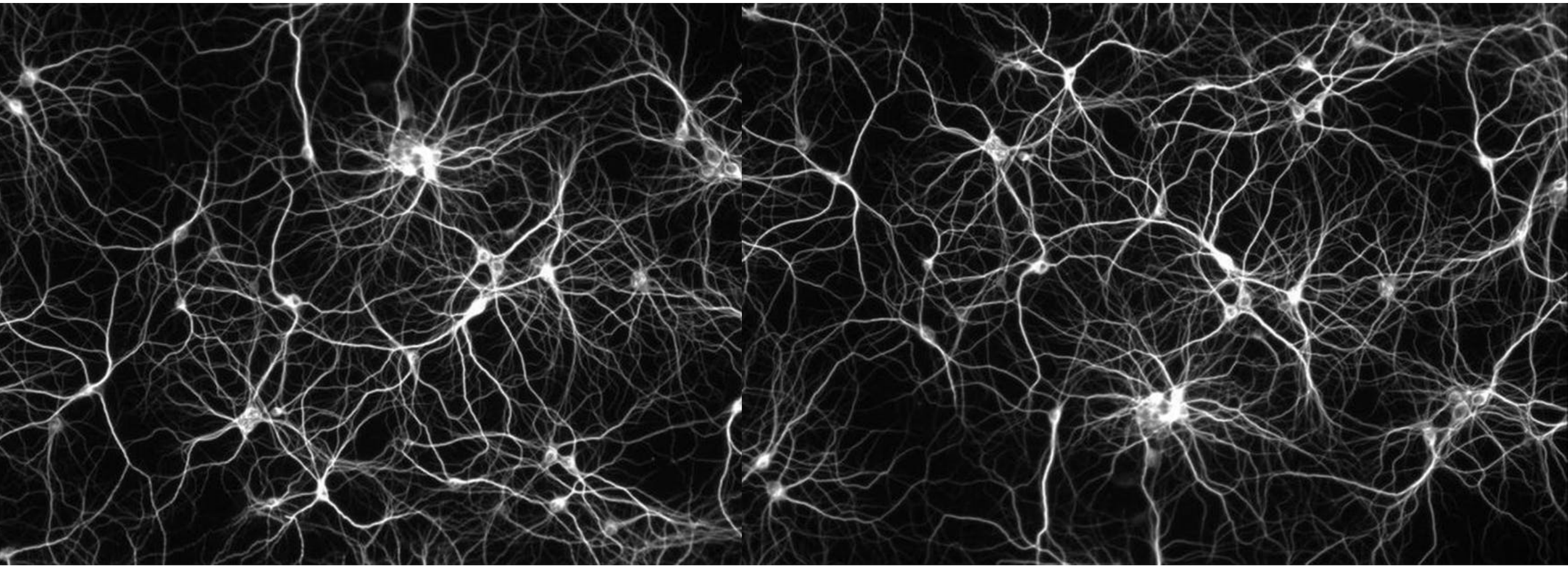
Learning in Deep Artificial and Biological Neuronal Networks

Lecture #2: Plasticity in the Brain.



ETH Zürich

Lecture starts at 9:15am

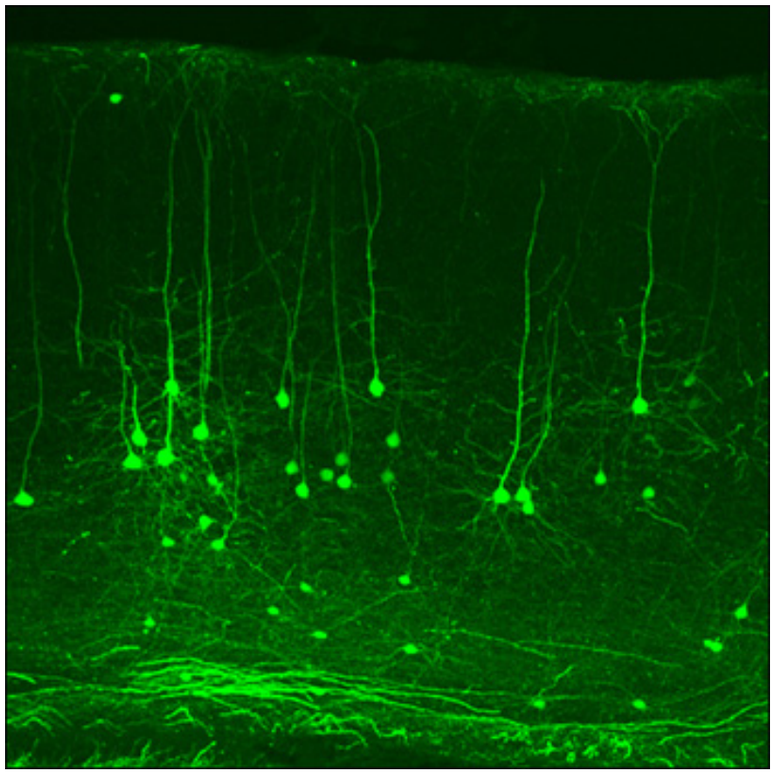
University of Zurich[UZH]

Benjamin F. Grewe
Institute of Neuroinformatics.
bgrewe@ethz.ch
Wednesday, Sept. 23

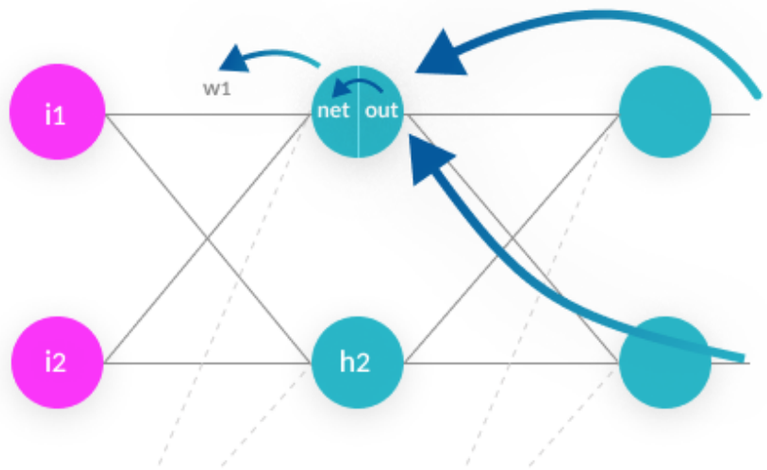
Plasticity in the Brain - Why is this Topic important?

- Biological plasticity might provide a different angle to understand the training procedures in DNNs.
- Given the effectiveness of human brain learning bio-plasticity might provide inspiration/new ideas for improved DNN training algorithms.
- Understanding biological plasticity might help to better understand how (hierarchical) learning is organized in the brain (Learning&Memory).
- Many neural disorders such as dementia relate to a disturbance in neuronal plasticity that cause neuronal networks to become dysfunctional.

Learning and Plasticity



DNN Training



Plasticity in the Brain - Content Lecture #2.

1. Why do we need plasticity?
2. Synaptic Plasticity:
   - 2.1) Basics of synaptic plasticity
   - 2.2) Homeostatic Plasticity.
   - 2.3) Hebb's Idea and STDP.
   - 2.4) Heterosynaptic Plasticity.
3. The Hippocampus as a model system to study neural plasticity
   - 3.1) LTP and LTD induction in the Hippocampus.
   - 3.2) Molecular basis of synaptic plasticity.
4. Non-synaptic plasticity:
   - 4.1) Neuronal excitability and spike generation.

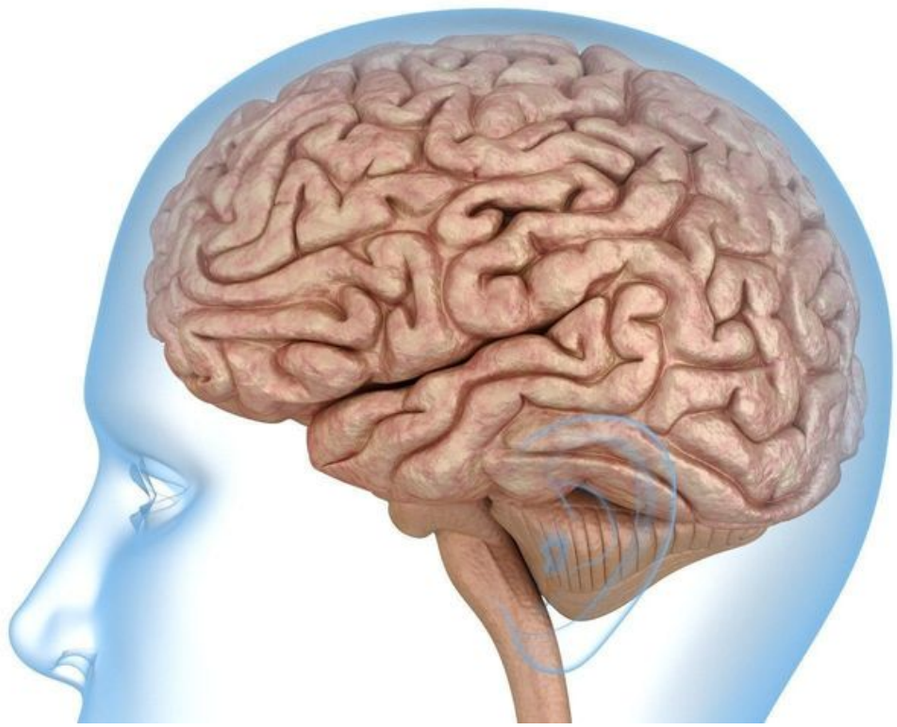
Goal of todays Lecture:
Provide an overview about how synaptic plasticity in the brain is organised.

The C. Elegans genome stores neuronal wiring!



The simple worm C. Elegans, for example, has 302 neurons and about 7000 synapses and in each individual of an inbred strain, the wiring pattern is exactly the same (Chen et al., 2006).

The Human Brain is mostly Learned.



The human brain has about 10^{11} neurons, and more than 10^{3} synapses per neuron. Specifying a connection target requires about $\log_2 10^{11} = 35$ bits/synapse. Thus it would take about 3.5×10^{15} bits (~400 TB) to specify all 10^{14} connections in the brain.

Learning Intelligent Behaviors Requires Plasticity.



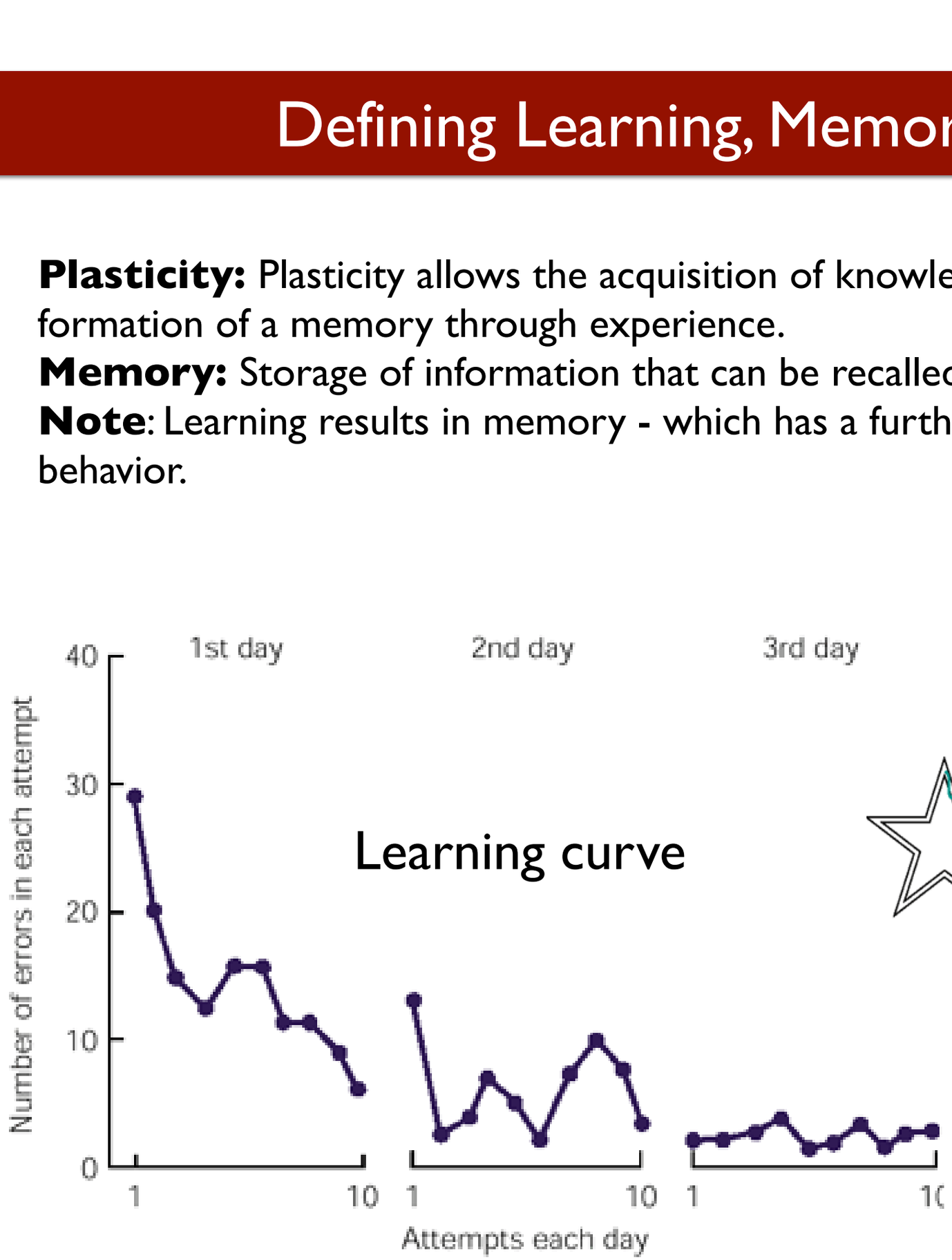




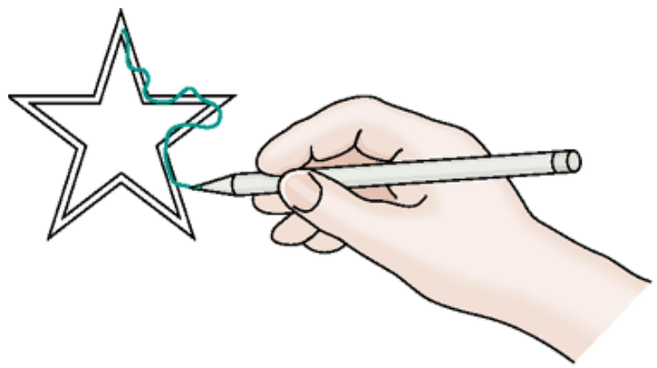


Defining Learning, Memory and Plasticity.

Plasticity: Plasticity allows the acquisition of knowledge/information and the formation of a memory through experience.
Memory: Storage of information that can be recalled at a later stage in time.
Note: Learning results in memory - which has a further outcome - it can change future behavior.



Learning curve

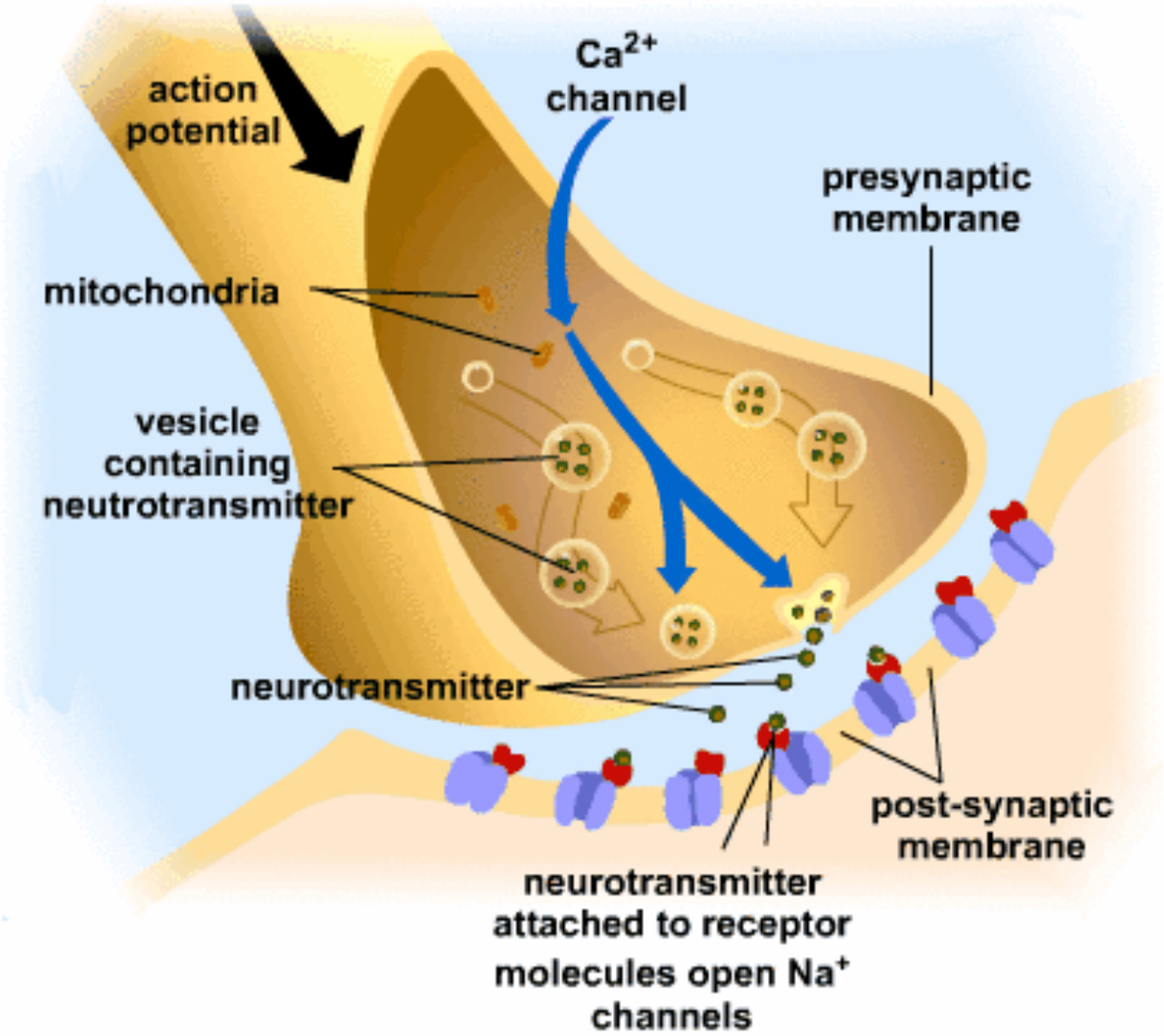


Kandel, *Principles of neural science*, 2001

The Definition of Neuronal Plasticity.

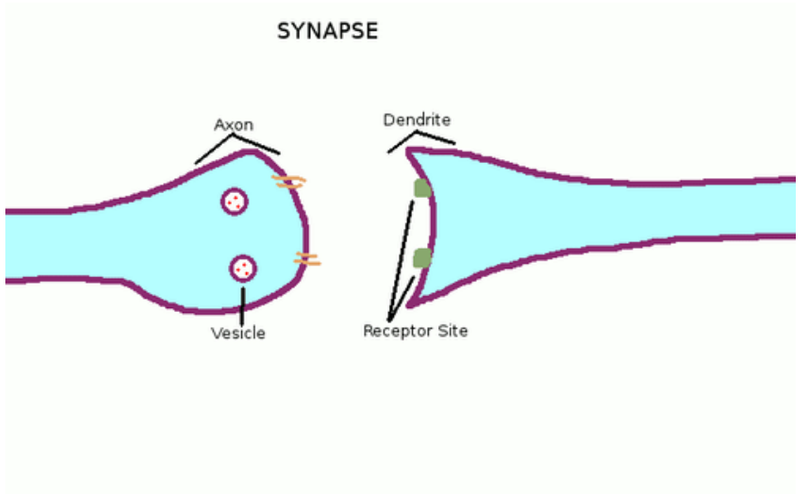
LTP: Long Term Potentiation
LTD: Long Term Depression

Synaptic Plasticity - A Short Recap of Synaptic Function.

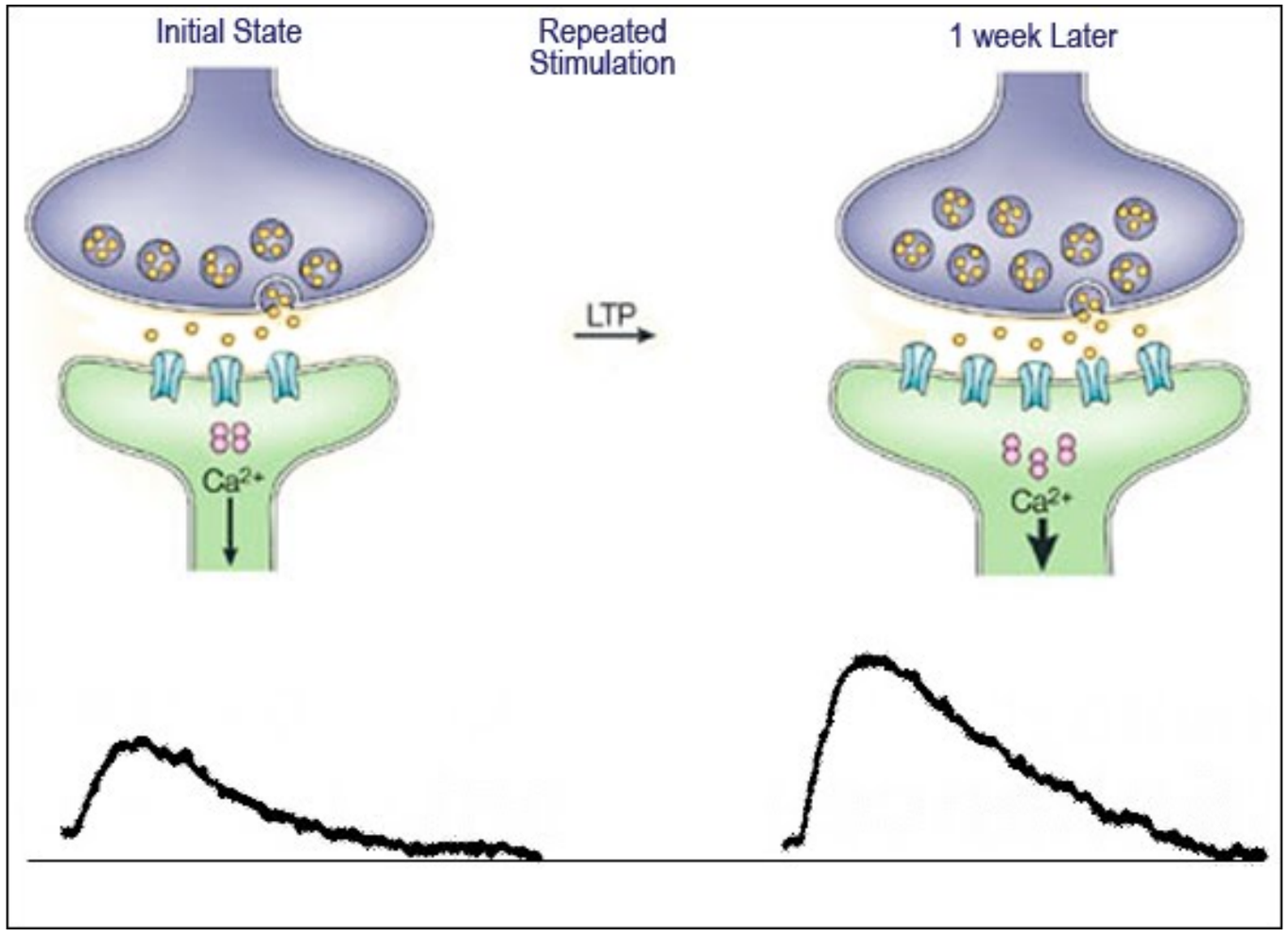


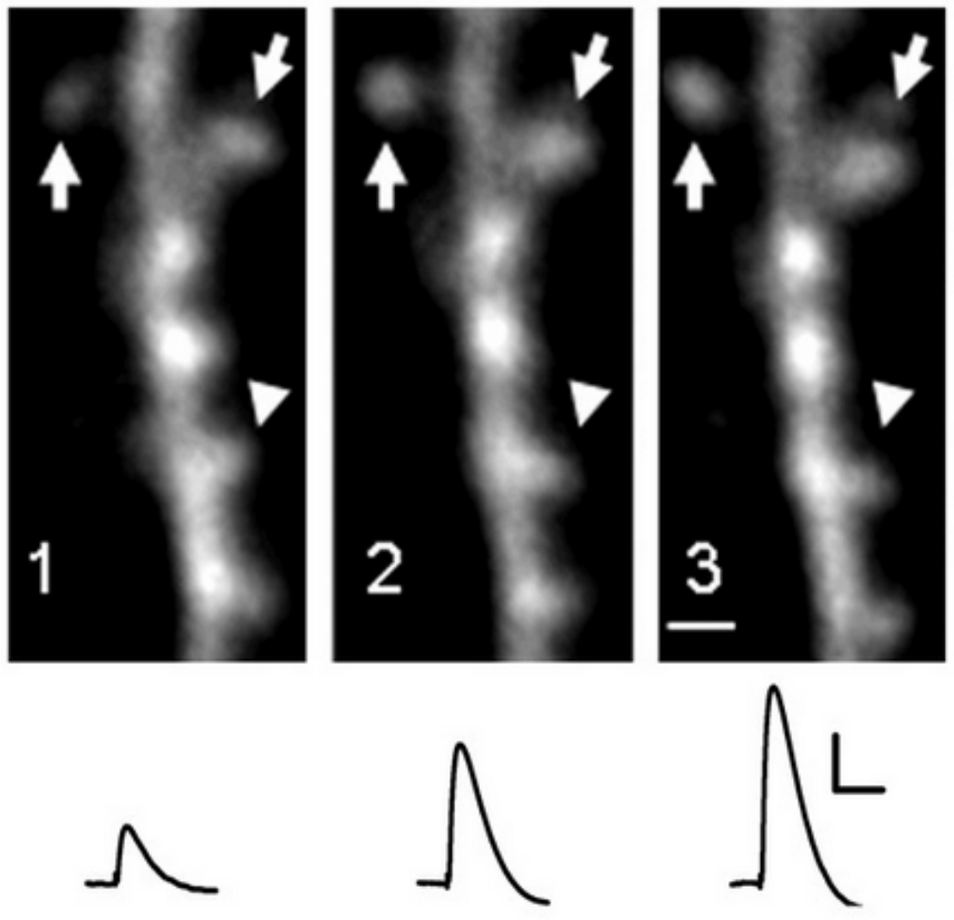
Neurotransmitters:

- Amino Acids …Glutamate, GABA
- Biogenic Amines … Dopamine, Histamine
- Neuropeptides … LHRH, Proctolin



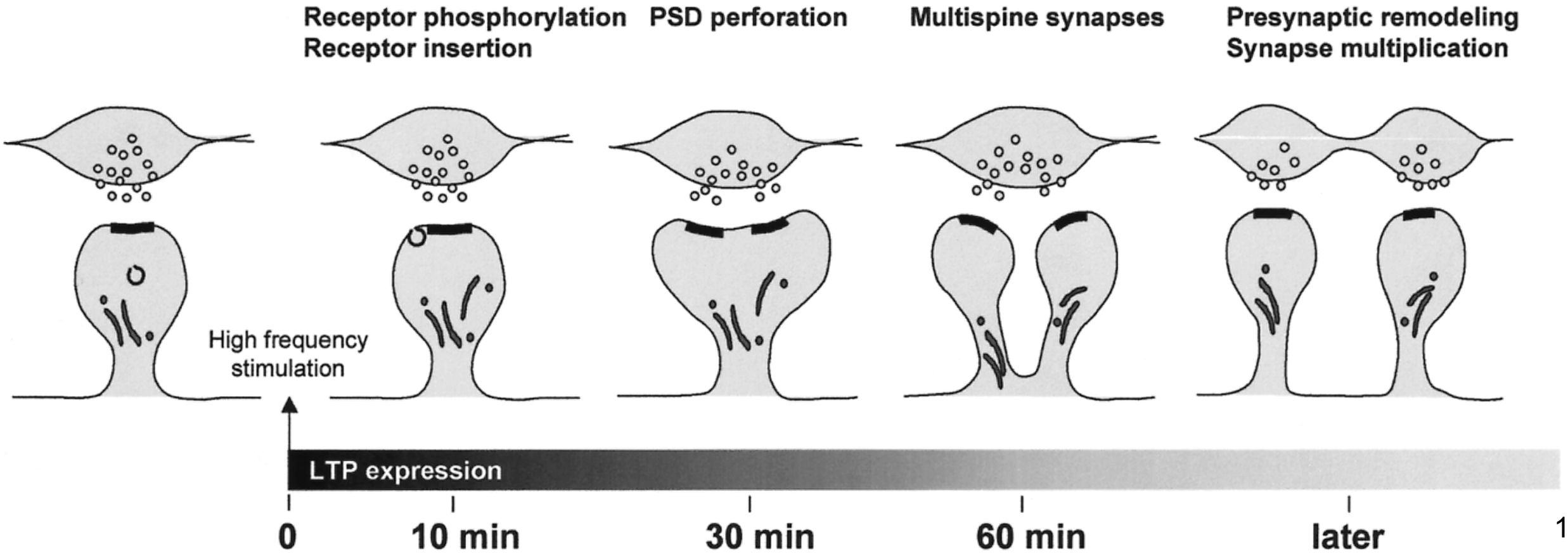
Synaptic Plasticity Alters Inter-neuron Connection Strength.





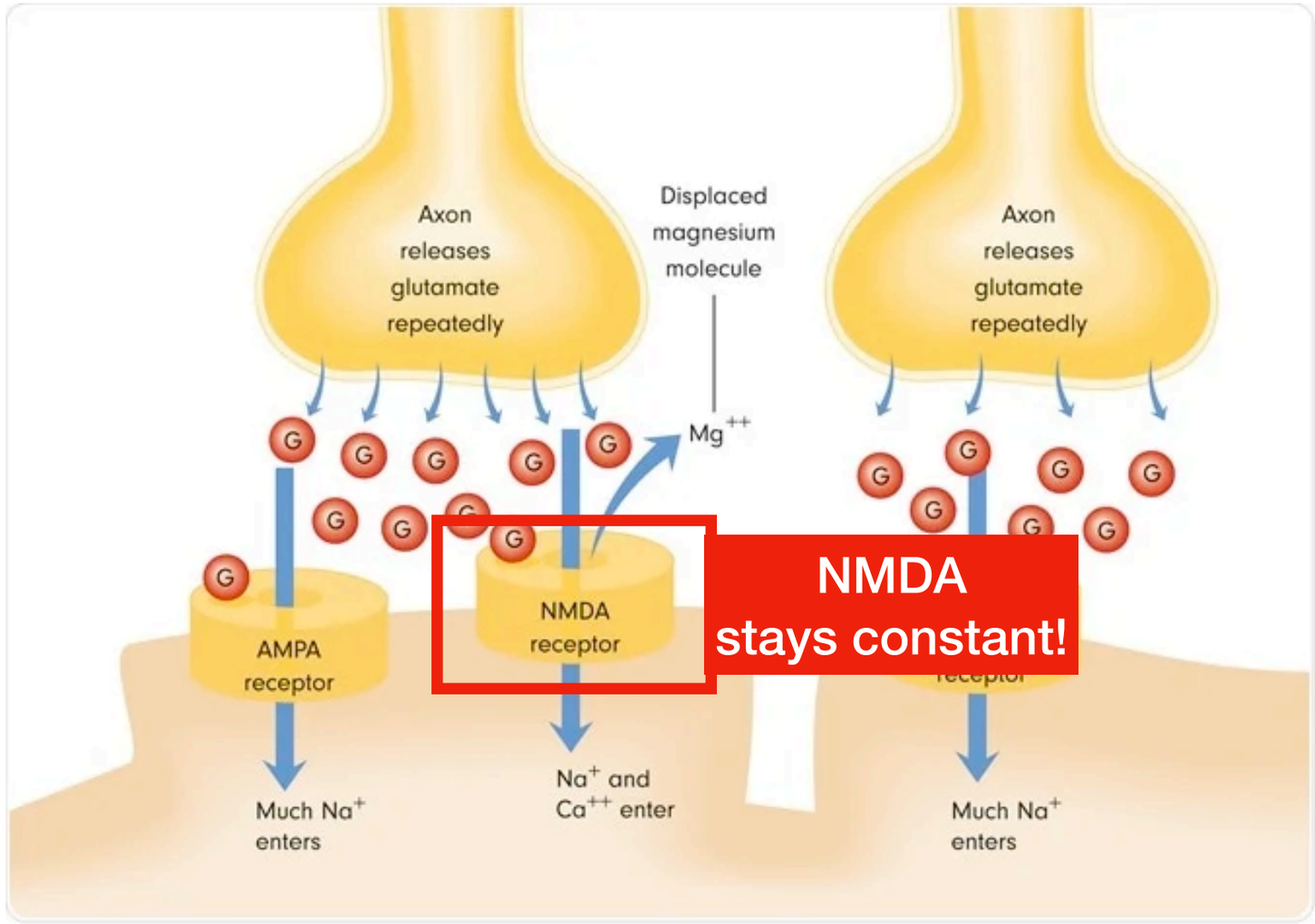
Synaptic Plasticity Alters the Inter-neuron Connection Strength.

1. Synaptic density size changes (short term).
2. AMPA/NMDA ratio changes (short term).
3. Number of spines changes (long term).

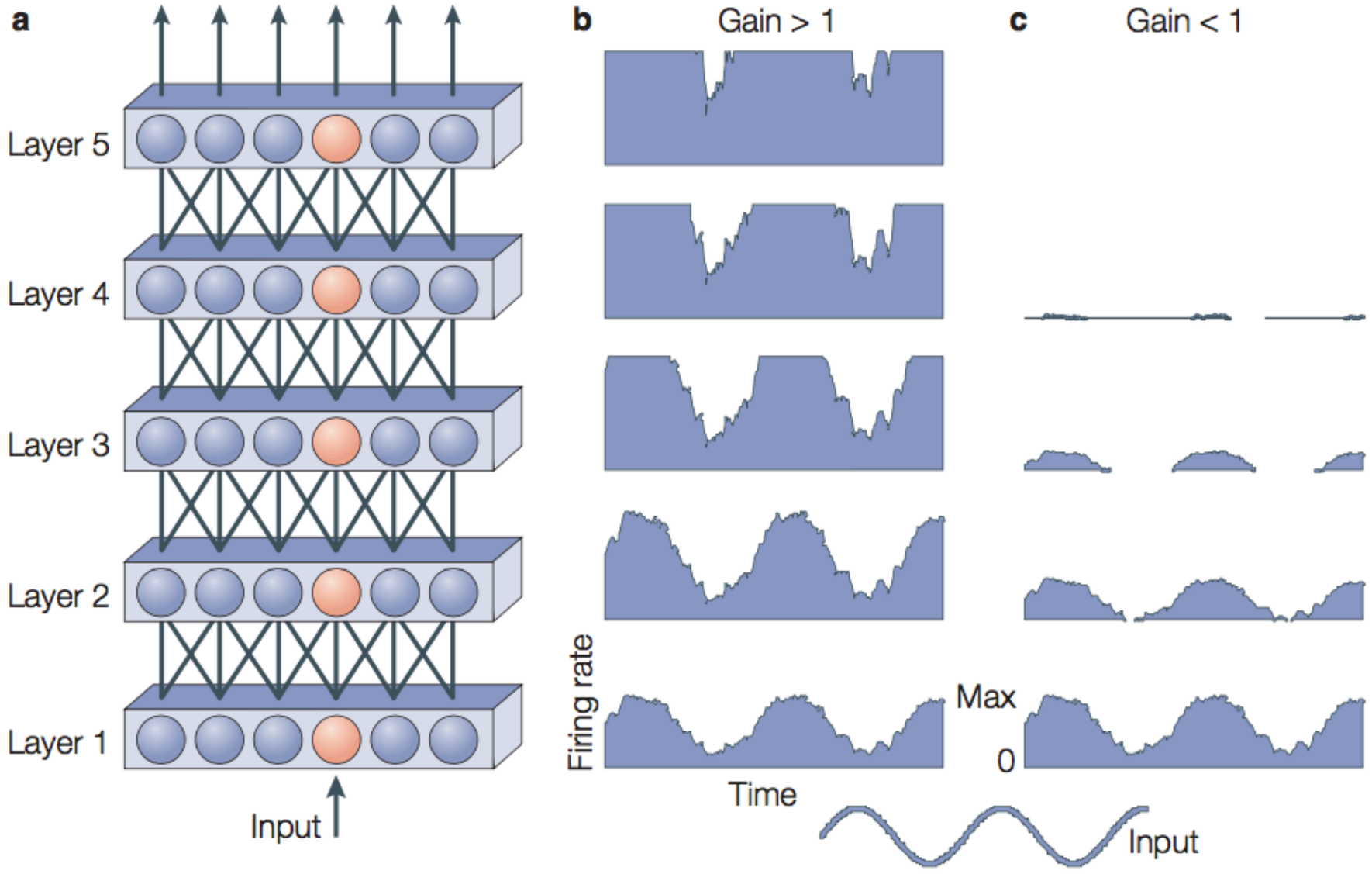


Synaptic Plasticity Alters the Inter-neuron Connection Strength.

1. Synaptic density size changes (short term).
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Necessity of Homeostatic Plasticity.



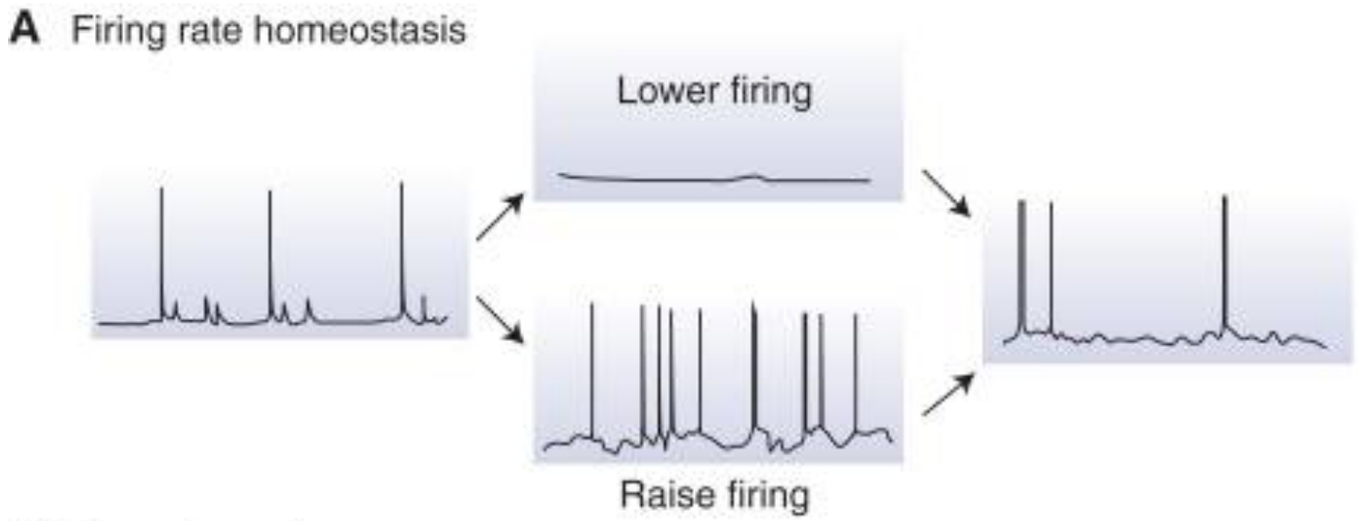
Review Article | Published: 01 February 2004

Homeostatic plasticity in the developing nervous system

Gina G. Turrigiano & Sacha B. Nelson

Nature Reviews Neuroscience **5**, 97–107 (2004) | Download Citation
1684 Accesses | **1316** Citations | **6** Altmetric | Metrics »

Homeostatic Plasticity.



Homeostatic Synaptic Plasticity: Local and Global Mechanisms for Stabilizing Neuronal Function

Turrigiano, 2012,

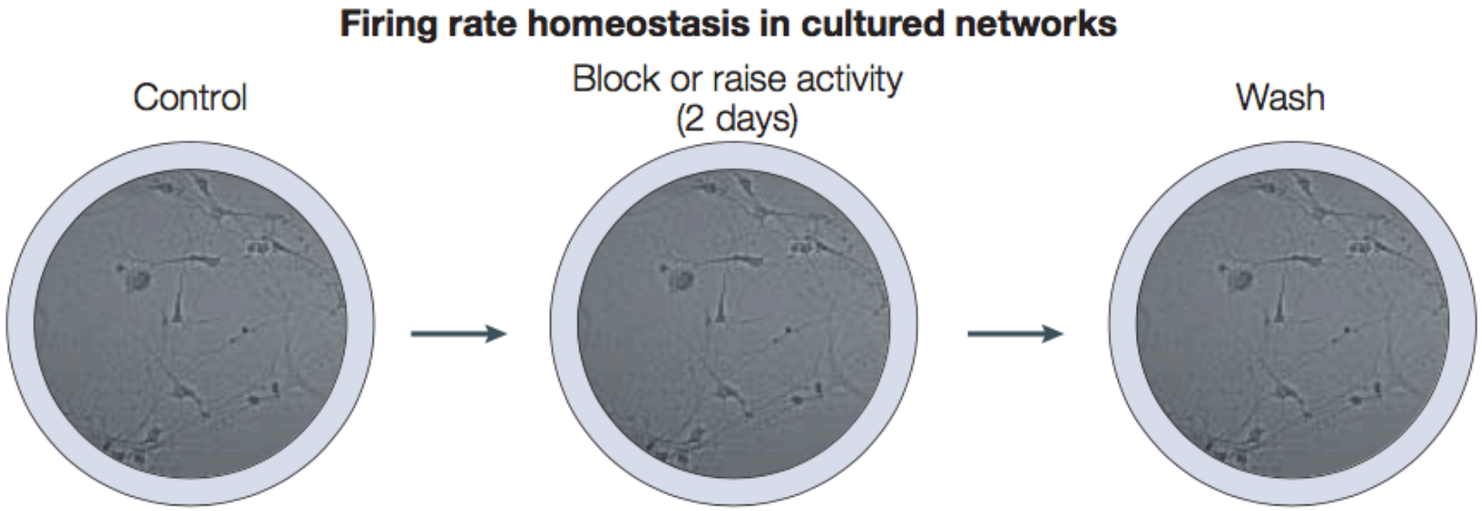
Gina Turrigiano

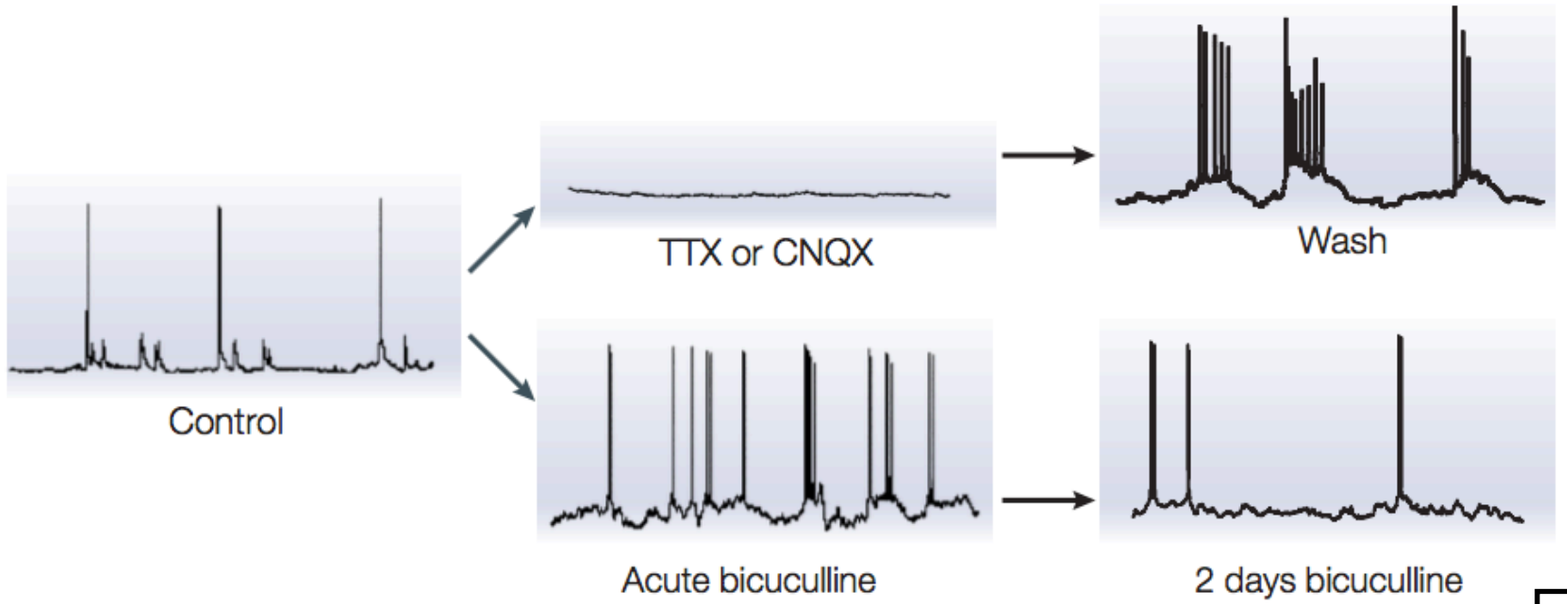
+ Author Affiliations

Correspondence: turrigiano@brandeis.edu

doi: 10.1101/cshperspect.a005736

Examples of Homeostatic Plasticity.





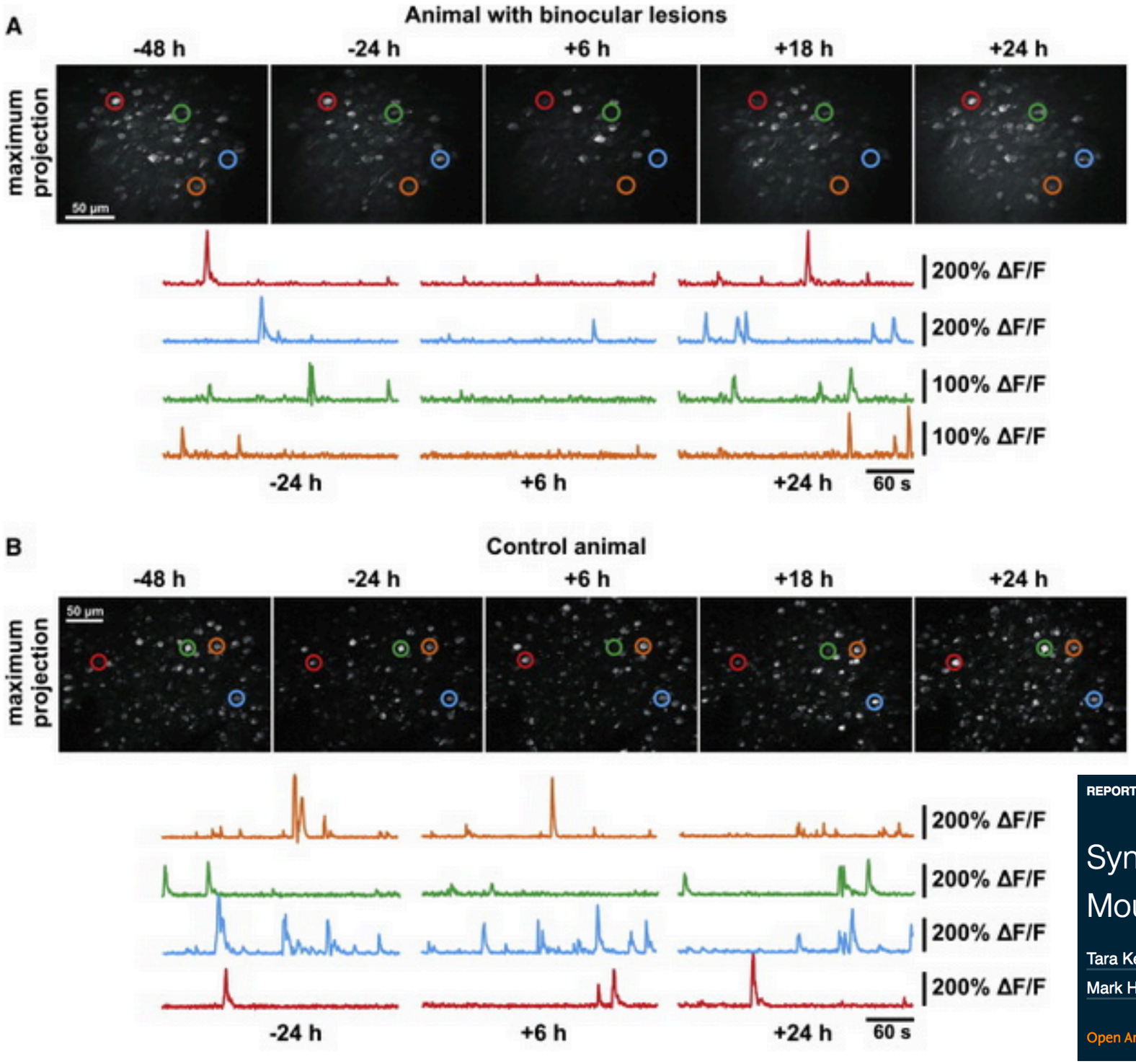
Review Article | Published: 01 February 2004

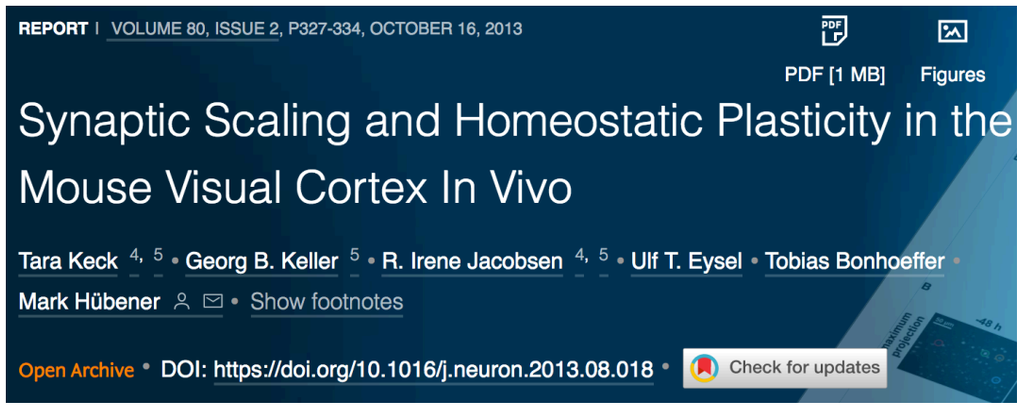
Homeostatic plasticity in the developing nervous system

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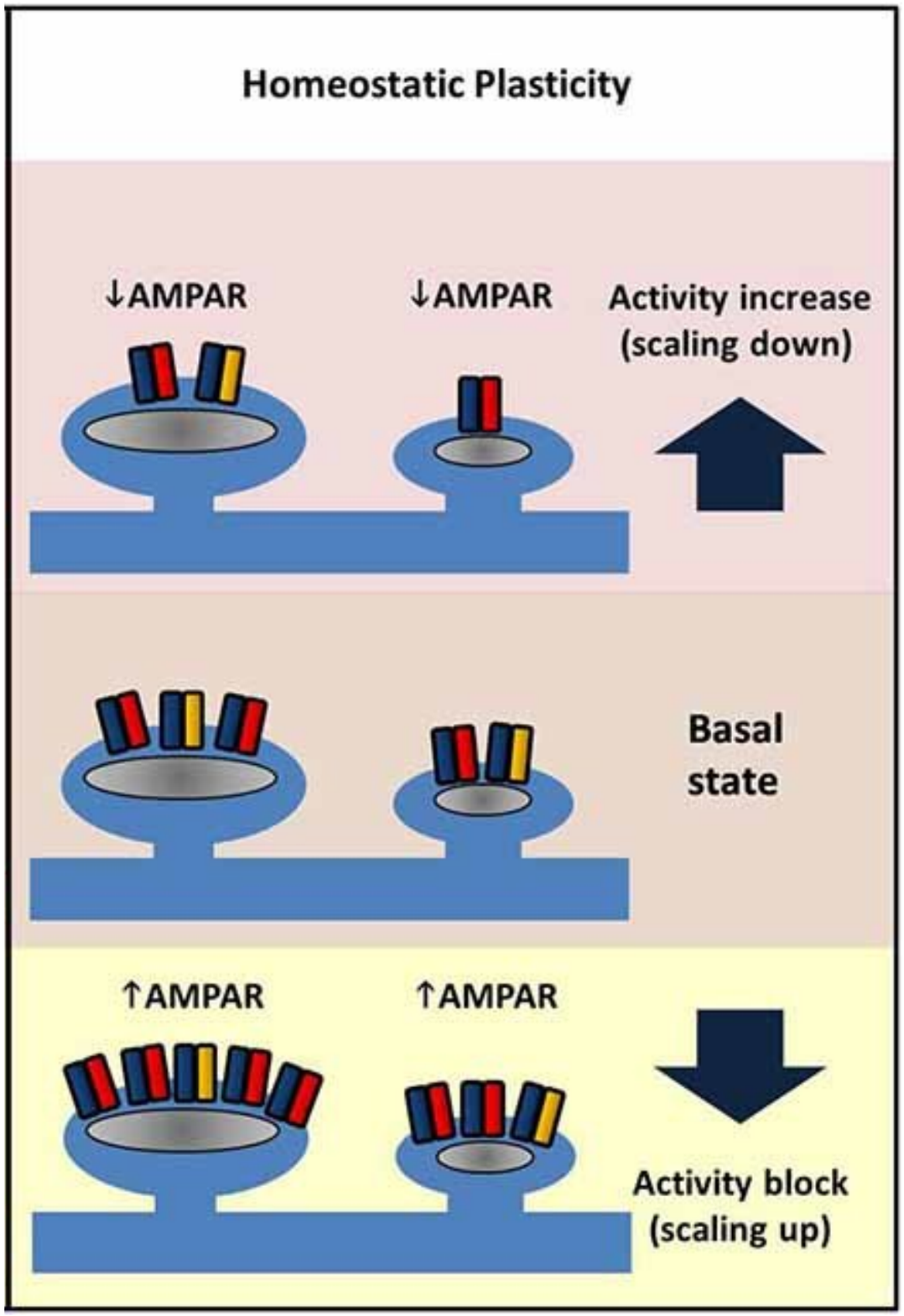
Nature Reviews Neuroscience **5**, 97–107 (2004) | Download Citation
1684 Accesses | **1316** Citations | **6** Altmetric | Metrics »

Examples of Homeostatic Plasticity.

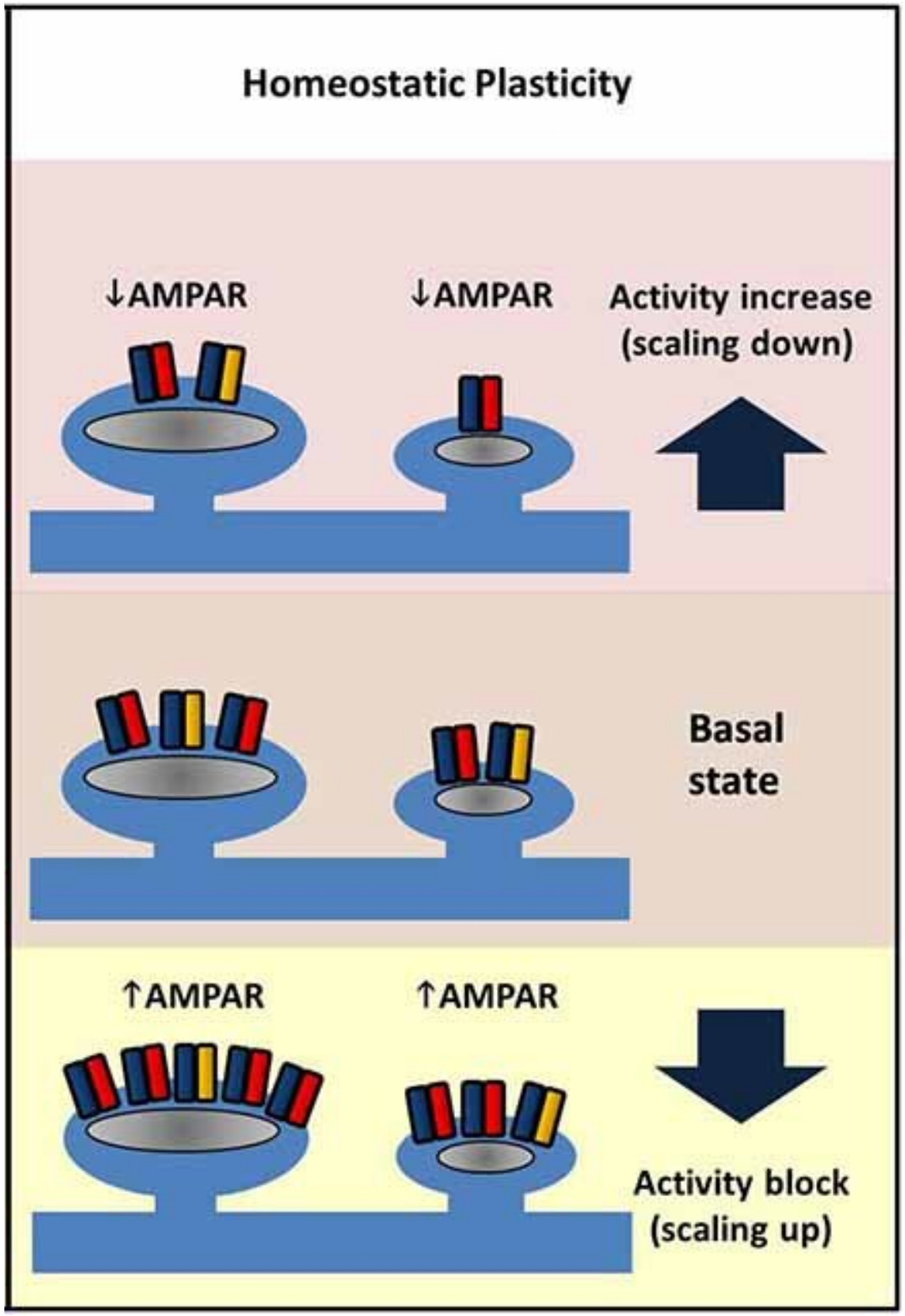


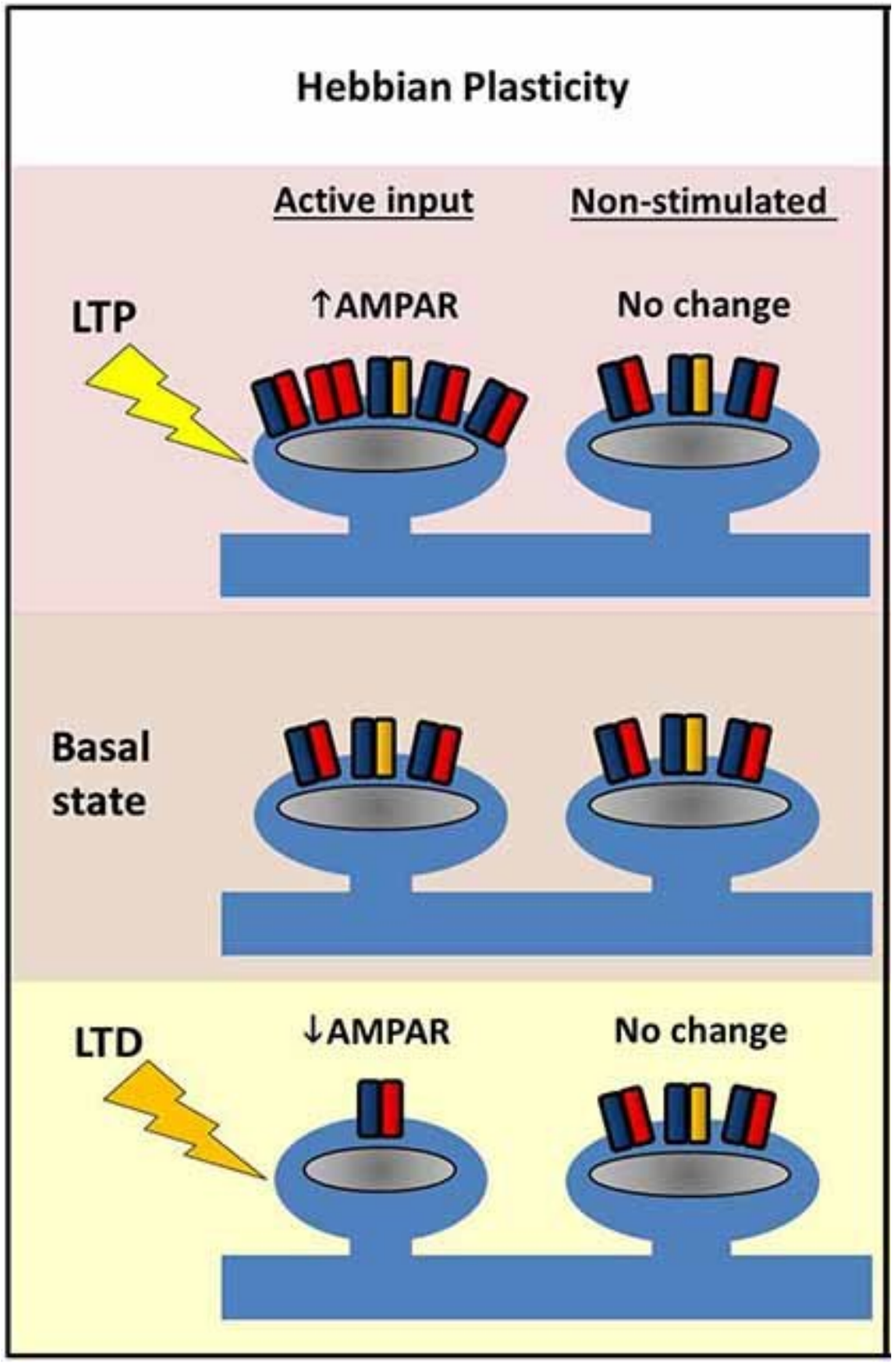


Homeostatic & Hebbian Plasticity.



Hebbian Learning - a Special Form of Homeostatic Plasticity.





The Hebbian Synapse.

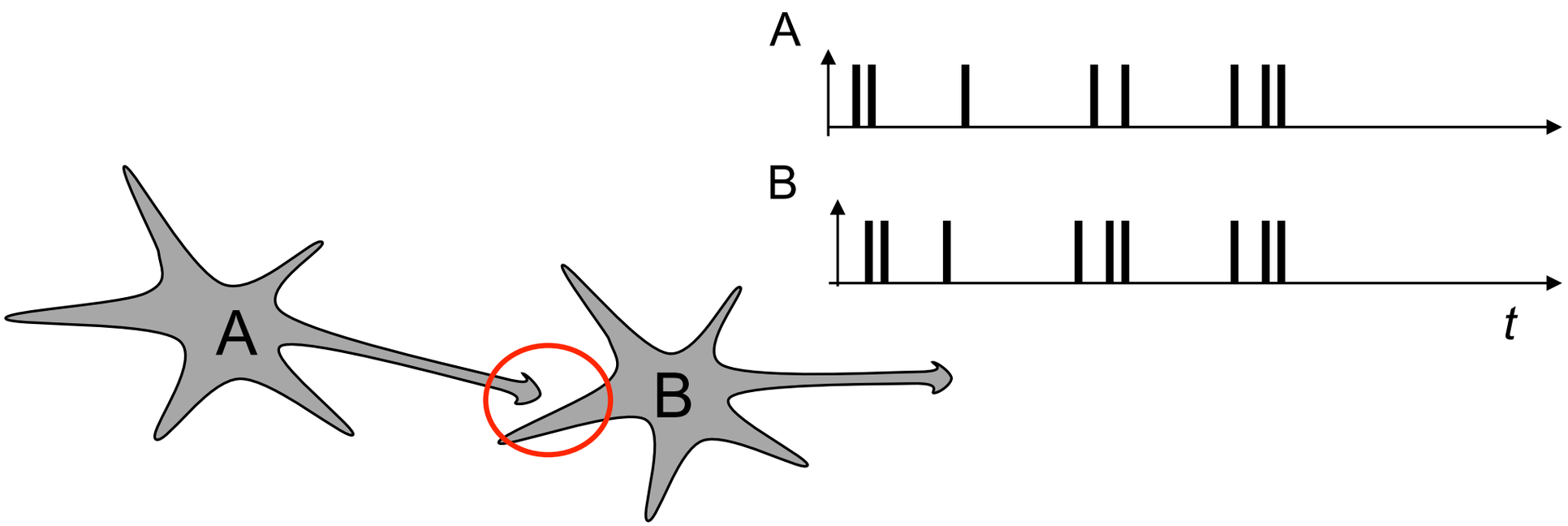


From *The Organization of Behavior* by Donald Hebb, 1949:

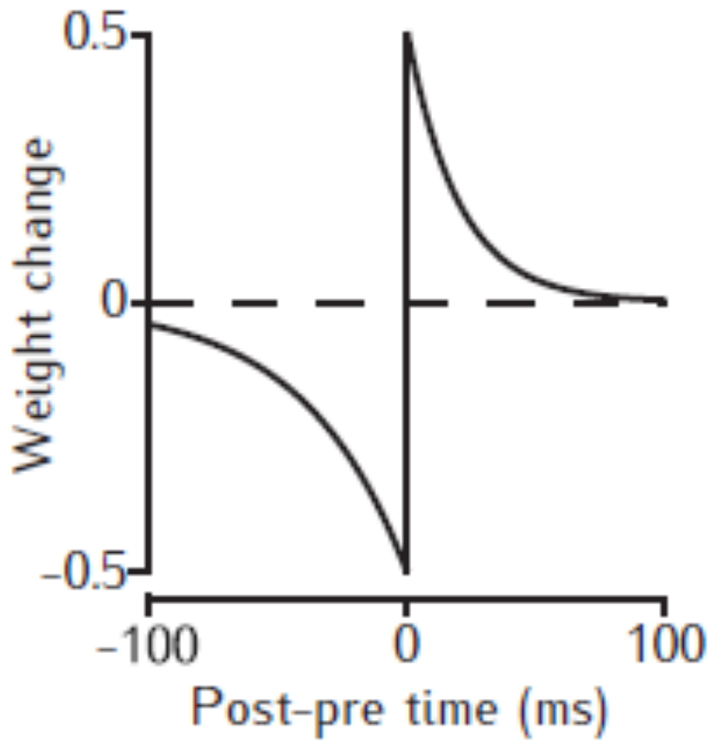
"When an axon of cell A is near enough to excite cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."

Hebb postulated that this behavior of synapses in neuronal networks would permit the networks to store memories.

A Hebbian synapse is a "coincidence detector"



Examples of HL - Spike-Timing Dependent Plasticity (STDP).



ARTICLE

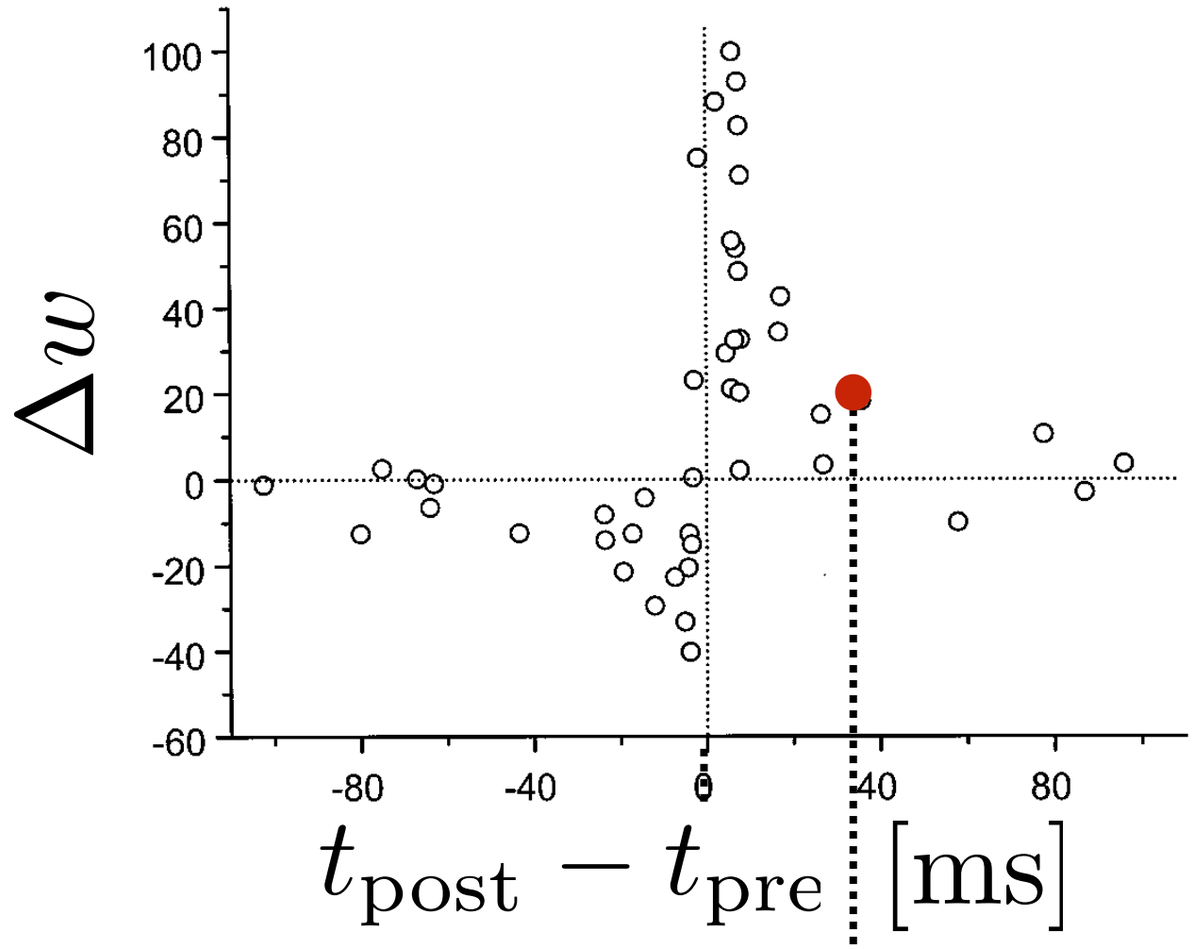
Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type

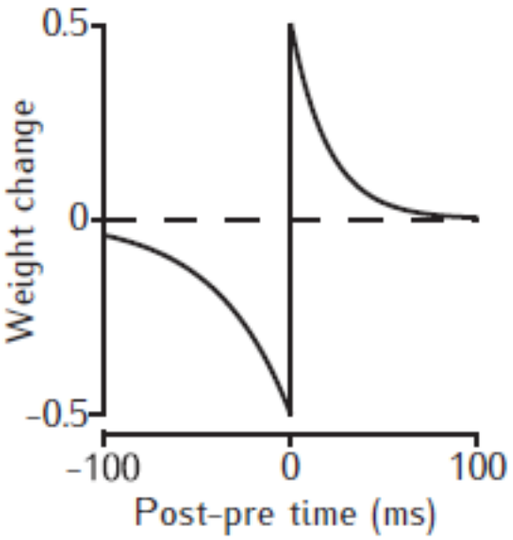
Guo-qiang Bi and Mu-ming Poo

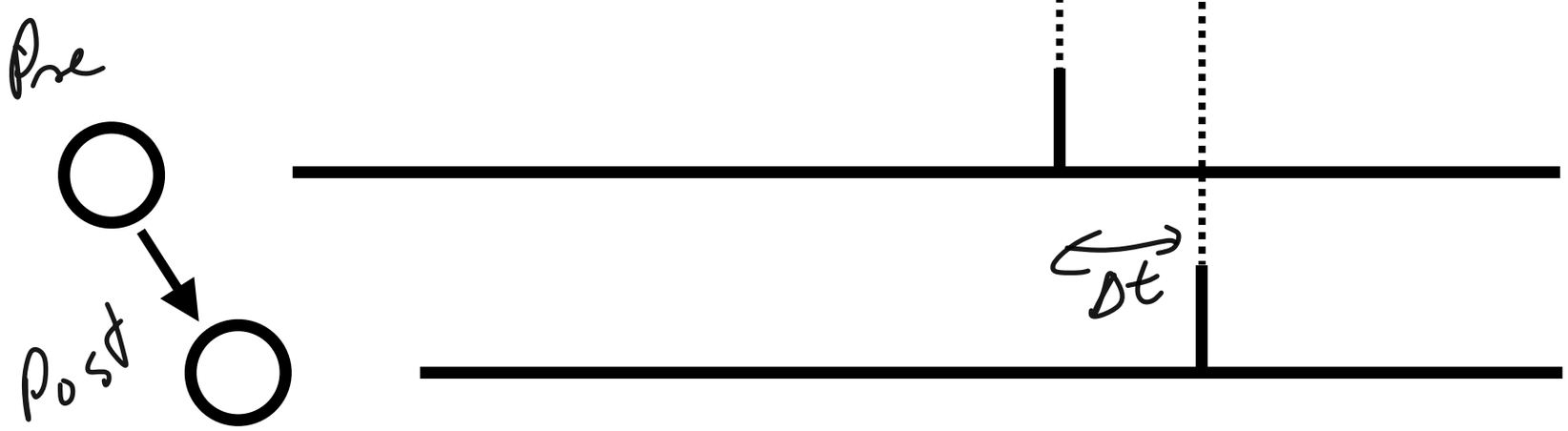
Journal of Neuroscience 15 December 1998, 18 (24) 10464-10472; DOI: https://doi.org/10.1523/JNEUROSCI.18-24-10464.1998

Examples of HL - Spike-Timing Dependent Plasticity (STDP).

Data

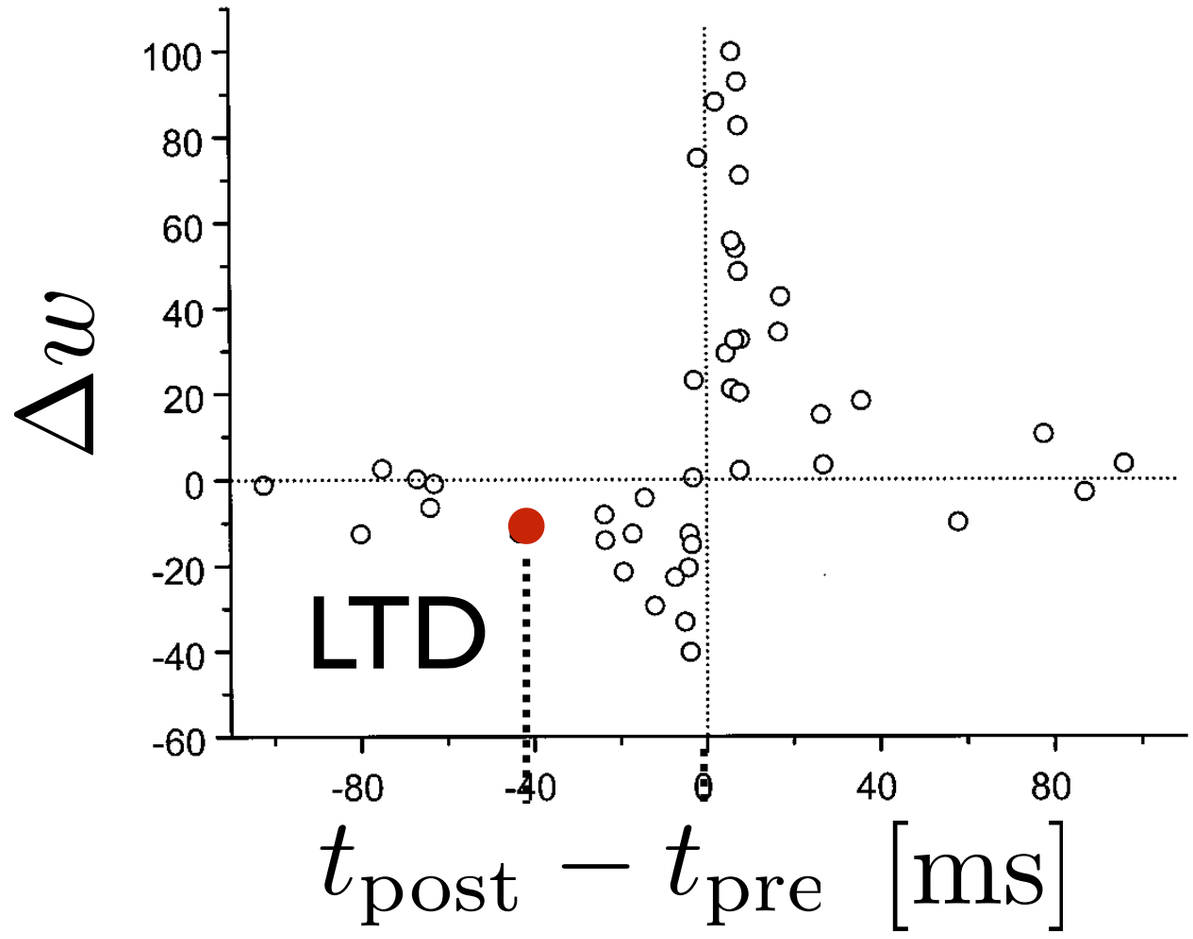


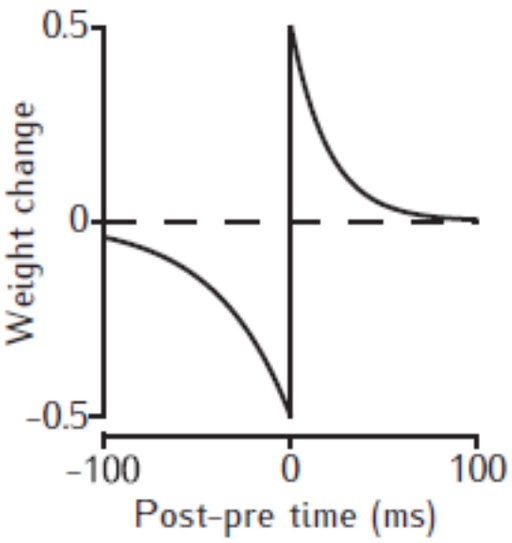


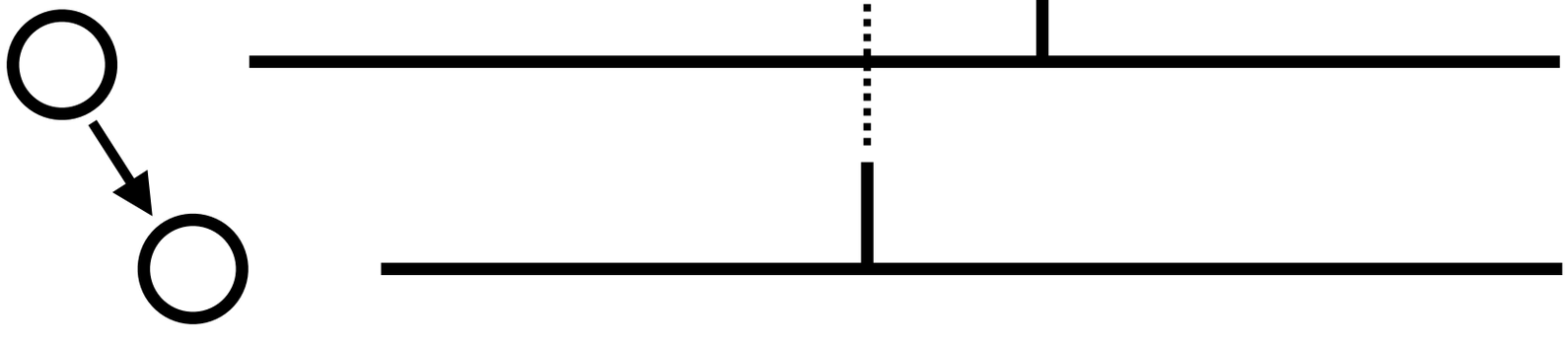


Examples of HL - Spike-Timing Dependent Plasticity (STDP).

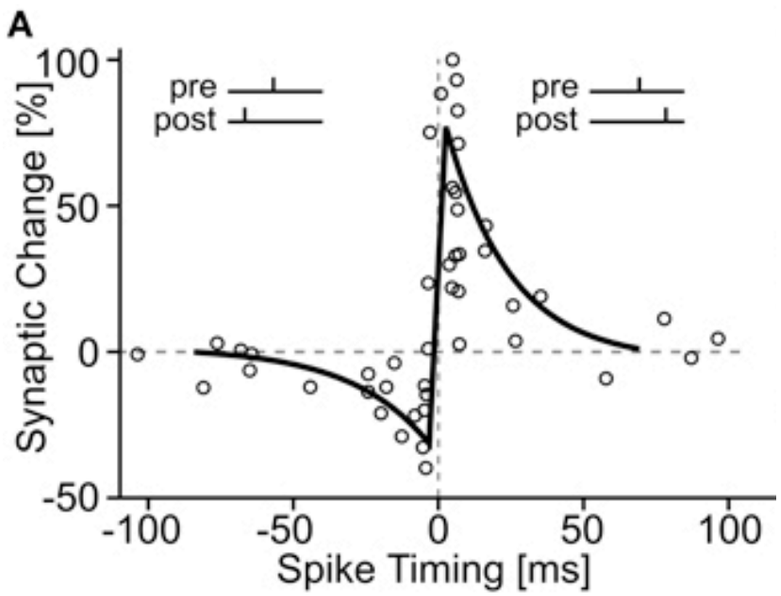
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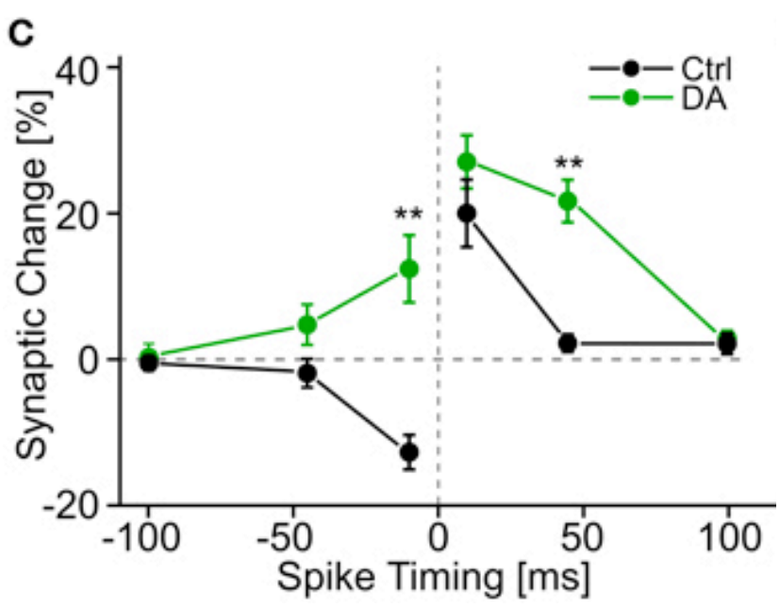






Neuromodulatory Effects in LTP.





ARTICLE

Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type

Guo-qiang Bi and Mu-ming Poo

Journal of Neuroscience 15 December 1998, 18 (24) 10464-10472; DOI: https://doi.org/10.1523/JNEUROSCI.18-24-10464.1998

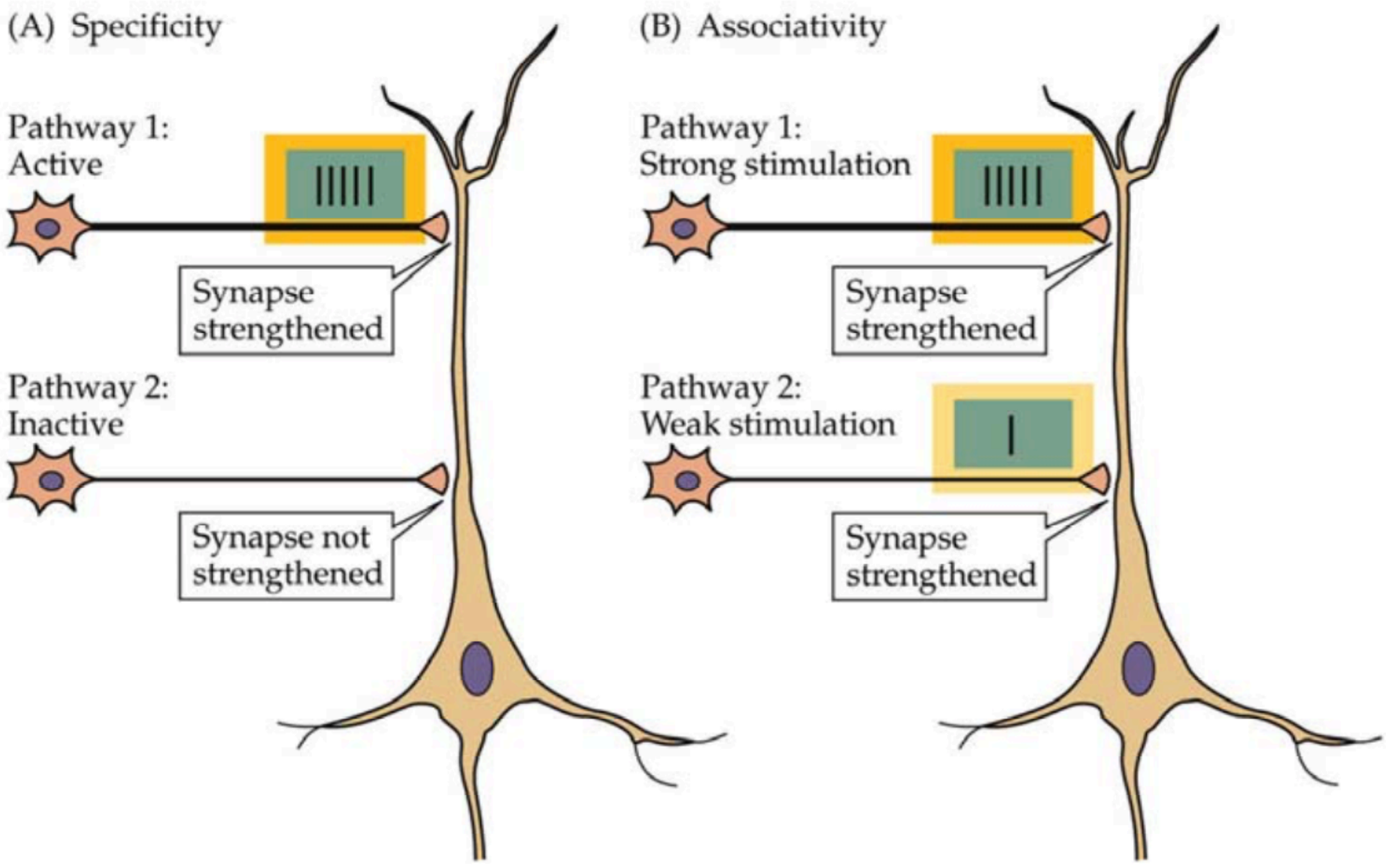
Gain in sensitivity and loss in temporal contrast of STDP by dopaminergic modulation at hippocampal synapses

Ji-Chuan Zhang, Pak-Ming Lau, and Guo-Qiang Bi

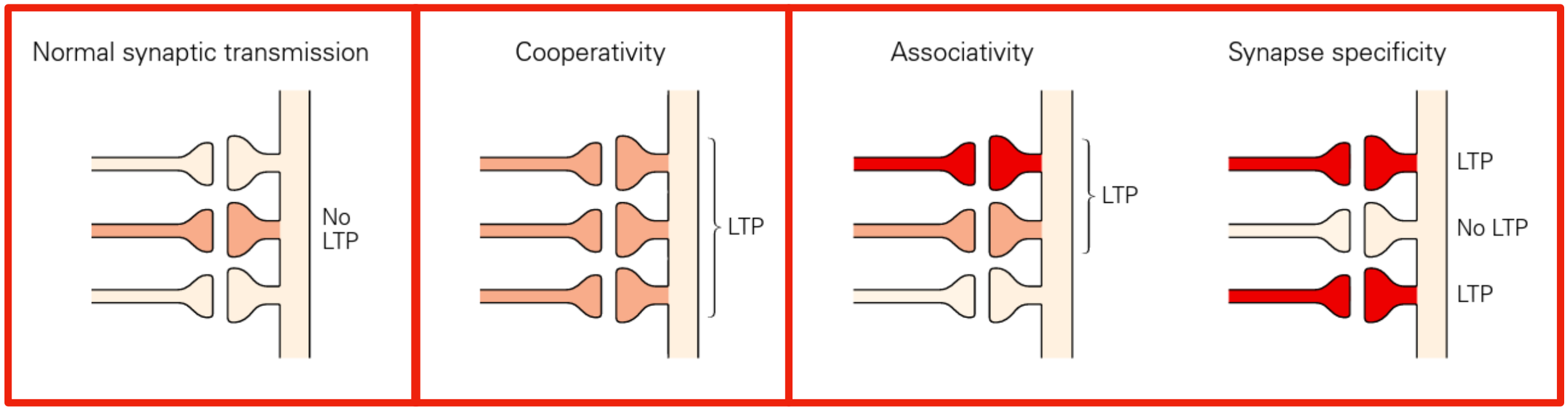
PNAS August 4, 2009 106 (31) 13028-13033; https://doi.org/10.1073/pnas.0900546106

The effect of the presence of neuromodulators *after* the pairing is not known.

Hebb's Idea how Neurons can Learn Associations.



Hebbian LTD and LTP are input specific.

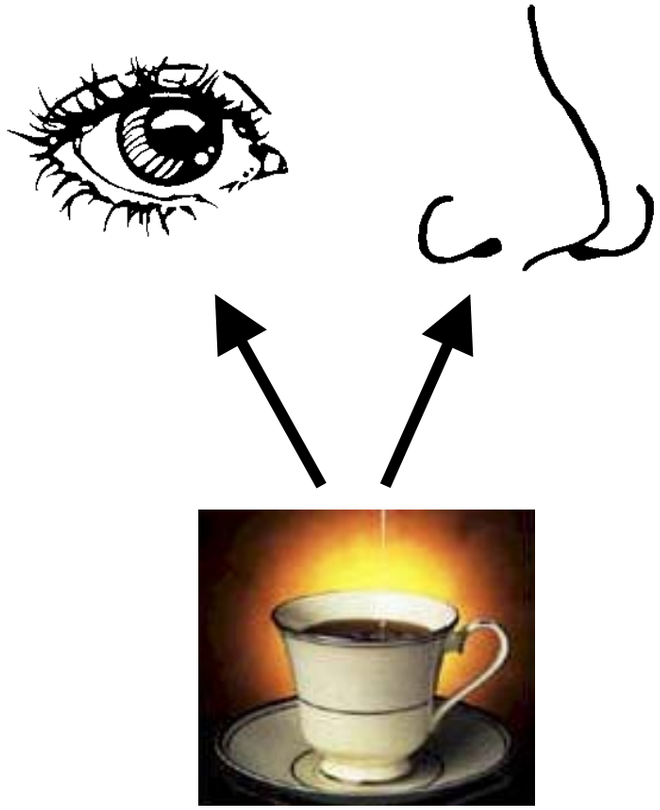


1. Cooperativity (induction threshold).
2. Input/synapse specificity.
3. Enables associative learning.

Hebb's Idea how Neurons can Learn Associations.



$$\dot{w} = H\,(\text{pre}, \text{post})$$



ARTICLE

doi:10.1038/nature21682

Neural ensemble dynamics underlying a long-term associative memory

Benjamin F. Grewe[1,2,3], Jan Gründemann[4], Lacey J. Kitch[1,2,3], Jerome A. Lecoq[1,2,3], Jones G. Parker[3,5], Jesse D. Marshall[1,2,3], Margaret C. Larkin[1,3], Pablo E. Jercog[1,2,3], Francois Grenier[4]†, Jin Zhong Li[1,3], Andreas Lüthi[4,6] & Mark J. Schnitzer[1,2,3]

The Problem with Hebbian Learning!

The Ups and Downs of Hebb Synapses

GEOFFREY HINTON
University of Toronto

Abstract

Modelers have come up with many different learning rules for neural networks. When a teacher specifies the correct output, error-driven rules work better than pure Hebb rules in which the changes in synapse strength depend on the correlation between pre and postsynaptic activities. But for unsupervised learning, Hebb rules can be very effective if they are combined with suitable normalization or "unlearning" terms to prevent the synapses growing without bound. Hebb rules that use rates of change of activity instead of activity itself are useful for discovering perceptual invariants and may also provide a way of implementing error-driven learning.

This approach led to effective "error-driven" learning rules such as the Widrow-Hoff rule (Widrow & Hoff, 1960) and the perceptron convergence procedure (Rosenblatt, 1961) and it was later generalized to multilayer networks by using backpropagation of the errors to get training signals for intermediate "hidden" layers (Rumelhart, Hinton, & Williams, 1986). Within the neural network community, the "Hebbian" approach of using the product of pre and postsynaptic activities to drive learning was seen as inferior to error-driven methods that use the product of the presynaptic activity and the postsynaptic activity *derivative* – the rate at which the objective function changes as the postsynaptic activity is changed. Even when the task was merely to associate random input vectors with random output vectors, it was shown that an error-driven rule worked much better than a Hebbian rule.

Unfortunately, error-driven learning has come...

What is Geoffrey's Problem with Hebbian Learning?

$$\dot{w} = H(\text{pre}, \text{post})$$

Hinton 2003, http://dx.doi.org/10.1037/

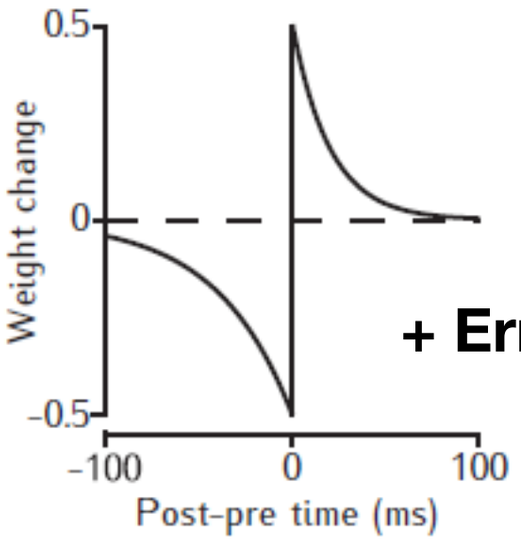
The Problem with Hebbian Learning!

What is Geoffrey's Problem with Hebbian Learning?

$$\dot{w} = H\,(\text{pre}, \text{post})$$

One Solution: Three Factor Hebbian Learning Rules!

$$\dot{w} = H\,(\text{pre}, \text{post})$$



+ Error term

$$\dot{w} = F\,(M, \text{pre}, \text{post})$$

Front Neural Circuits. 2015; 9: 85. PMCID: PMC4717313
Published online 2016 Jan 19. doi: 10.3389/fncir.2015.00085 PMID: 26834568

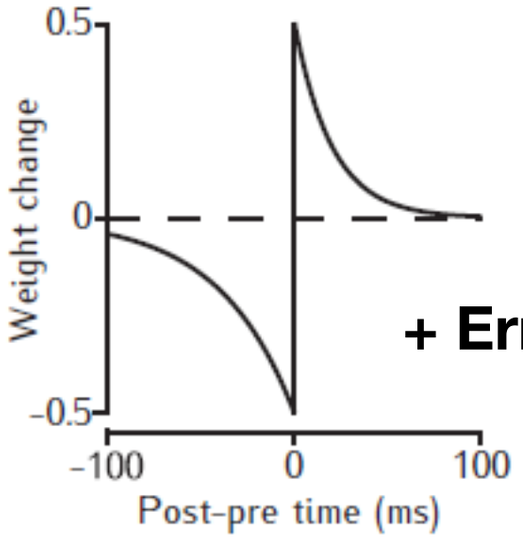
Neuromodulated Spike-Timing-Dependent Plasticity, and Theory of Three-Factor Learning Rules

Nicolas Frémaux and Wulfram Gerstner*

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One Solution: Three Factor Hebbian Learning Rules!

$$\dot{w} = H\,(\text{pre}, \text{post})$$



+ Error term

$$\dot{w} = F\,(M, \text{pre}, \text{post})$$

Front Neural Circuits. 2015; 9: 85. PMCID: PMC4717313
Published online 2016 Jan 19. doi: 10.3389/fncir.2015.00085 PMID: 26834568

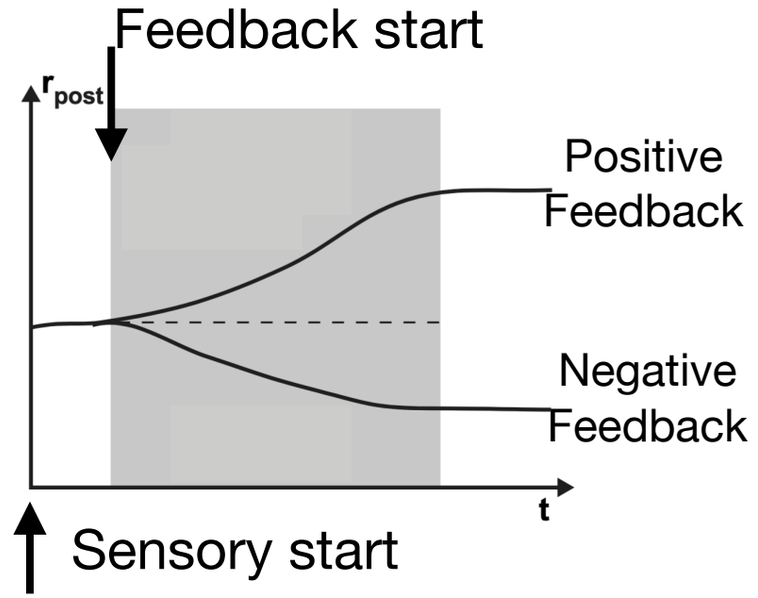
Neuromodulated Spike-Timing-Dependent Plasticity, and Theory of Three-Factor Learning Rules

Nicolas Frémaux and Wulfram Gerstner*

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Learning Through Control

Differential Hebbian + Feedback Control



Weight Update

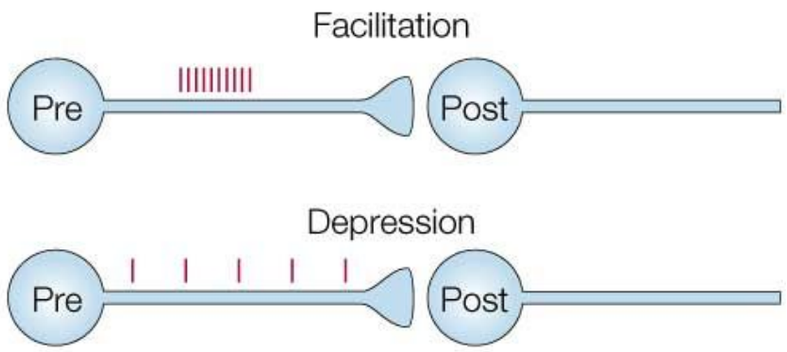
$$\Delta w \propto \int r_{\text{pre}}(t)\dot{r}_{\text{post}}(t)dt$$



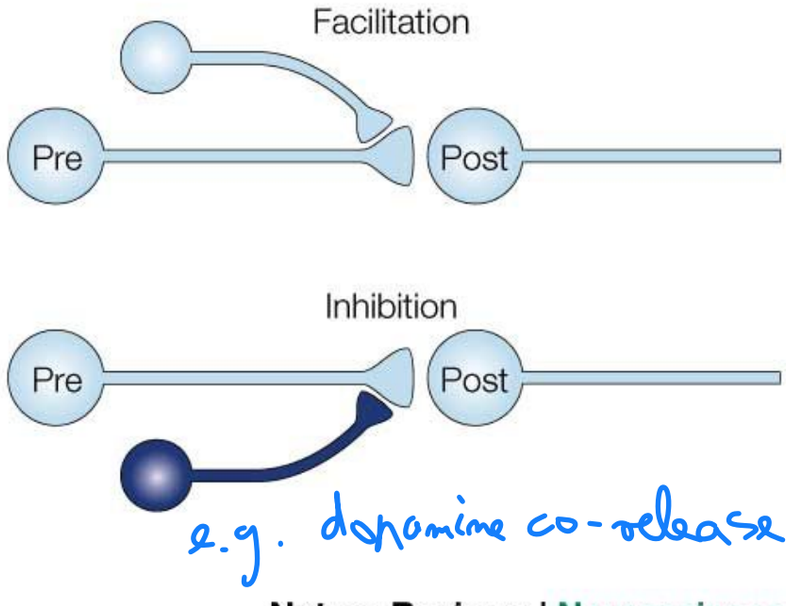
Aceituno et al, 2023, Front. Comp. Neurosc

Non-Hebbian Plasticity - Heterosynpatic Plasticity

a Homosynaptic (activity-dependent) plastic change



b Heterosynaptic (modulatory input-dependent) plastic change



Nature Reviews | Neuroscience

Review Article | Published: 01 October 2000

Is Heterosynaptic modulation essential for stabilizing hebbian plasiticity and memory

Craig H. Bailey, Maurizio Giustetto, Yan-You Huang, Robert D. Hawkins & Eric R. Kandel

Nature Reviews Neuroscience **1**, 11–20 (2000) | Download Citation ↓

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Consolidate your Memories!

Break!

Neuronal Plasticity in the Hippocampus.

The Nobel Prize in Physiology or Medicine 2014

John O'Keefe, May-Britt Moser, Edvard Moser

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The Nobel Prize in Physiology or Medicine 2014



Photo: David Bishop, UCL

John O'Keefe

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Photo: G. Mogen/NTNU

May-Britt Moser

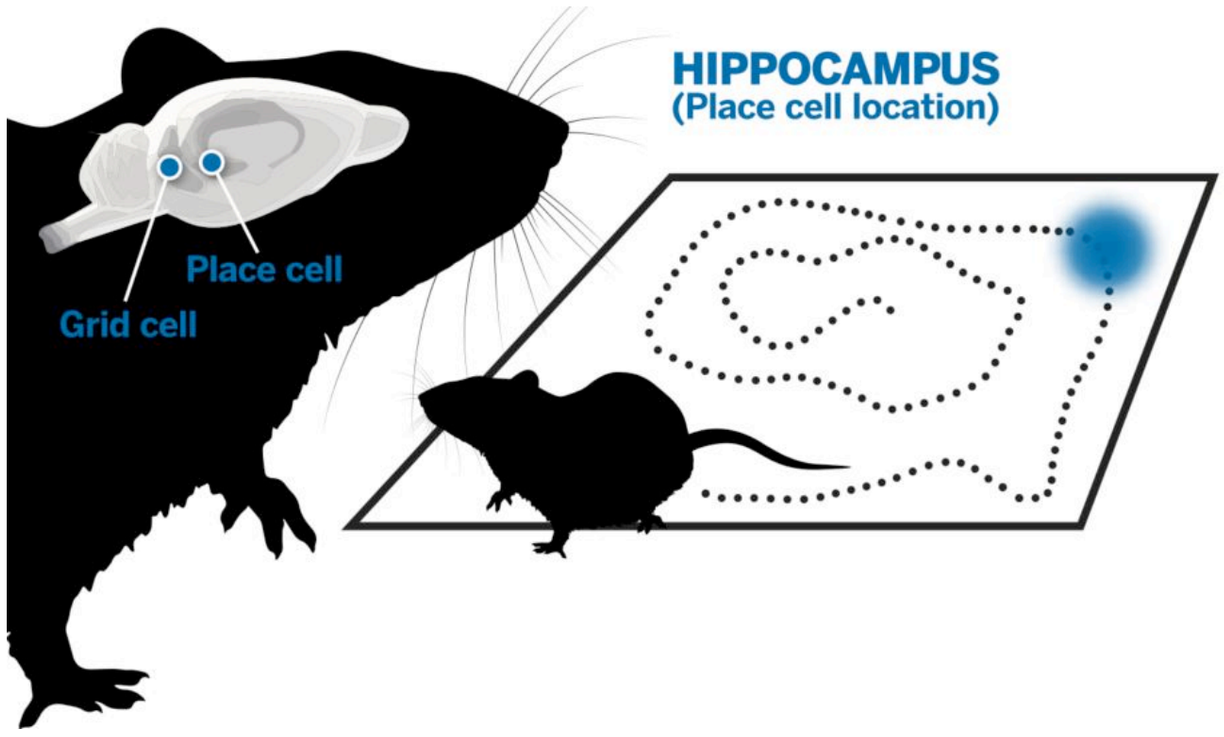
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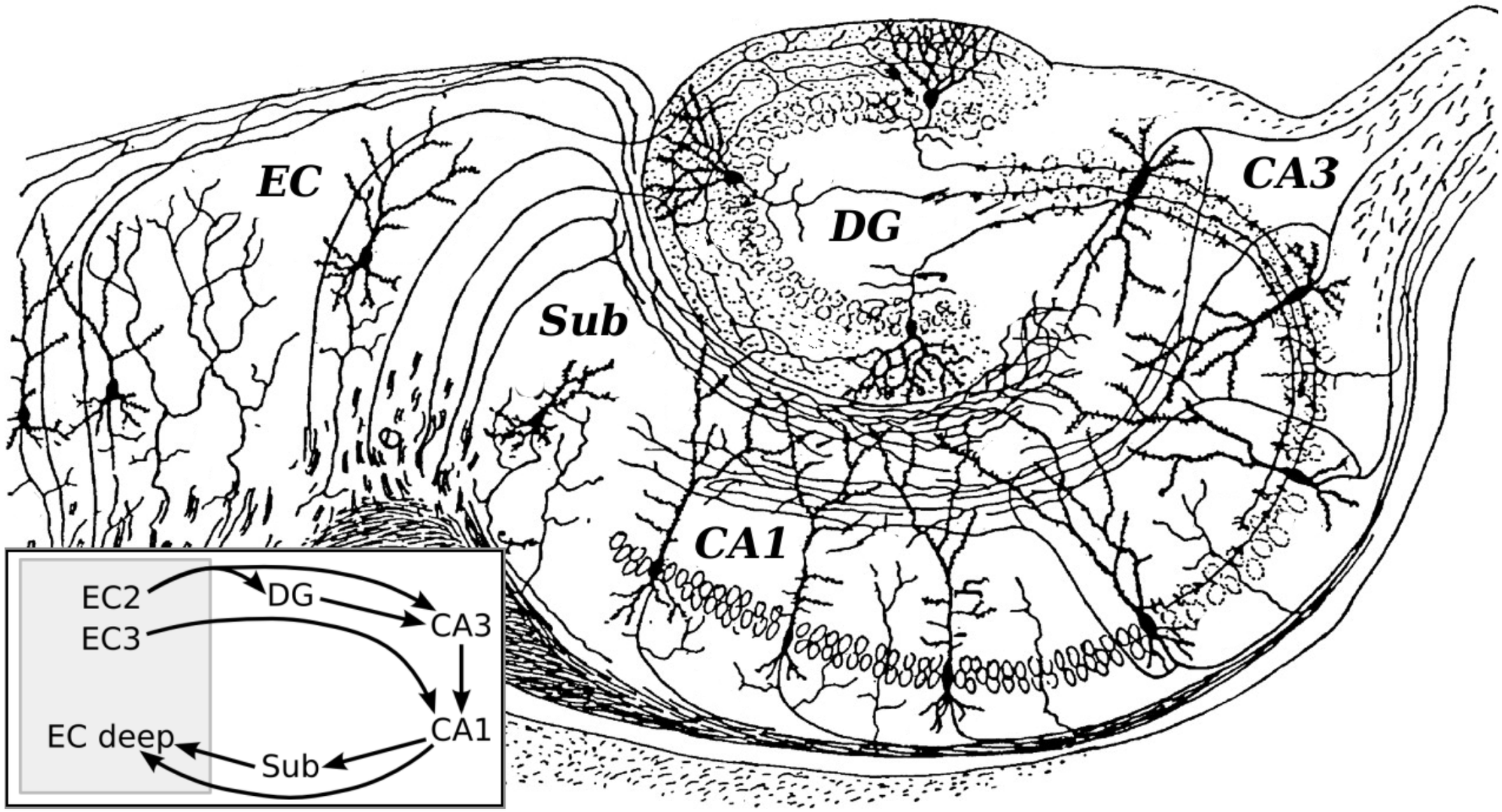
Photo: G. Mogen/NTNU

Edvard I. Moser

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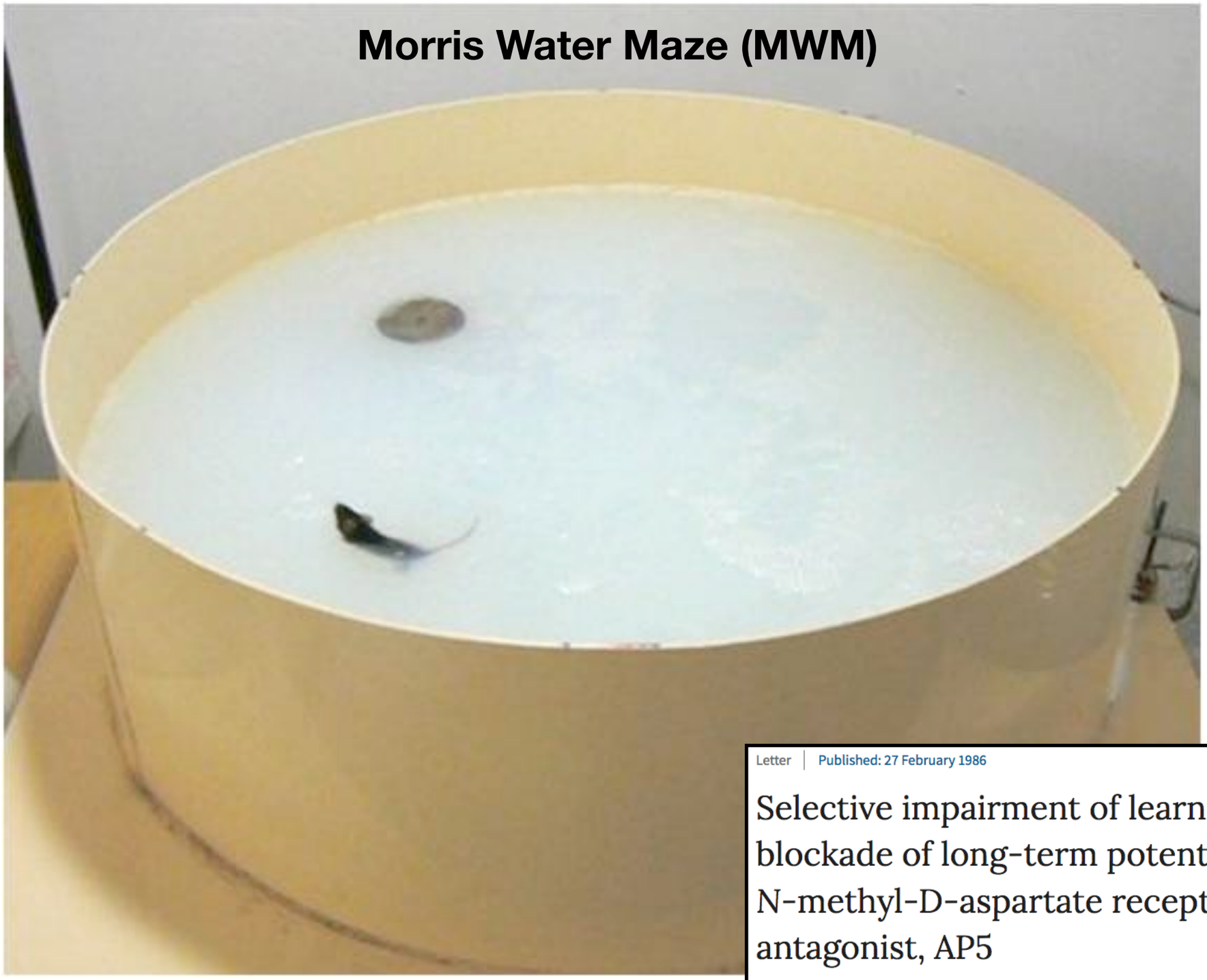


The Hippocampus as Model System to Study Plasticity.



A classic image from Ramon y Cajal illustrating the principle cells (excitatory projection cells) of the hippocampus (we won't deal with the numerous GABA interneurons). The DG has modified pyramidal cells (called granule cells) that have only apical dendrites. The CA fields contain fairly typical pyramidal cells with both basal and apical dendrites, both covered in spines.

The Hippocampus is Required for Spatial Learning.



Letter | Published: 27 February 1986

Selective impairment of learning and blockade of long-term potentiation by an N-methyl-D-aspartate receptor antagonist, AP5

R. G. M. Morris, E. Anderson, G. S. Lynch & M. Baudry

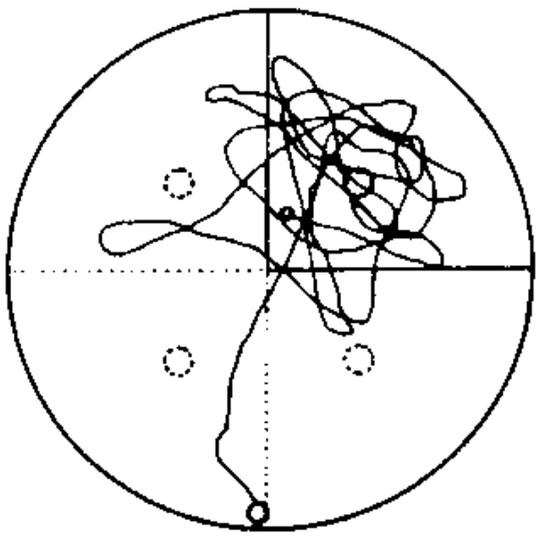
Nature **319**, 774–776 (1986) | Download Citation ↓

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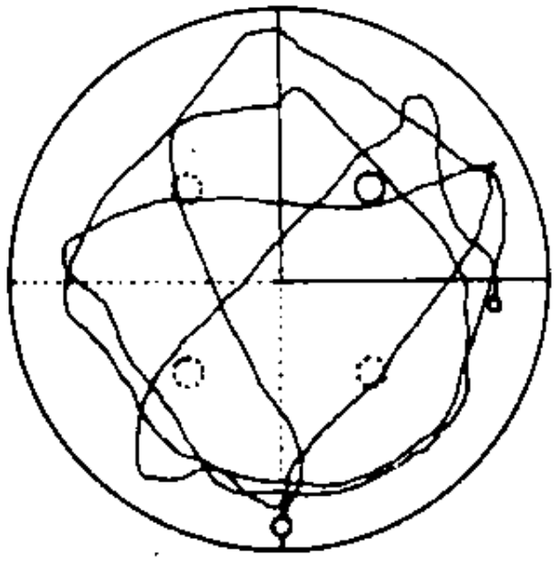
The Hippocampus is Required for Spatial Learning.

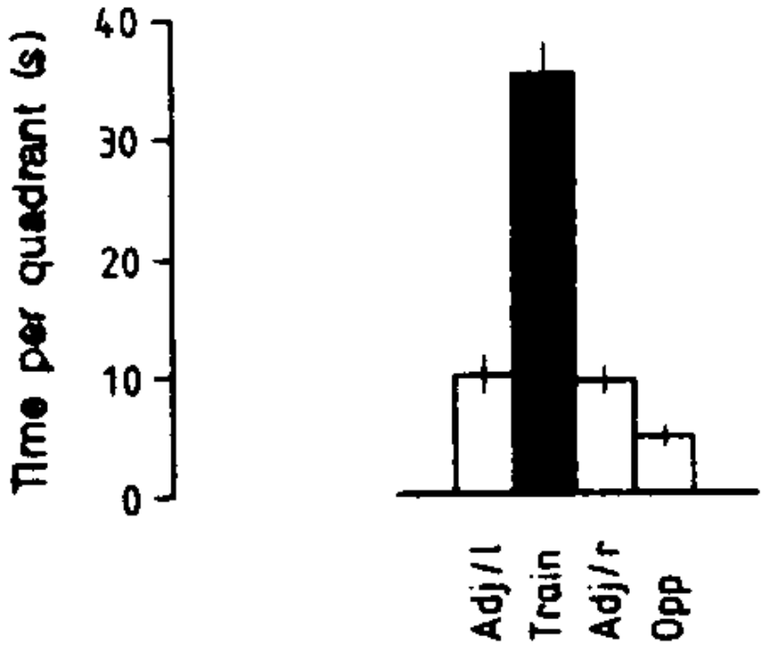


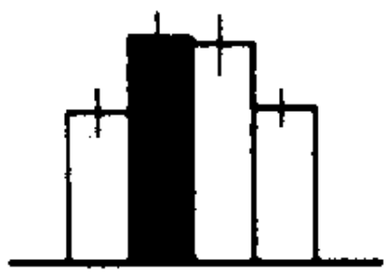
Control



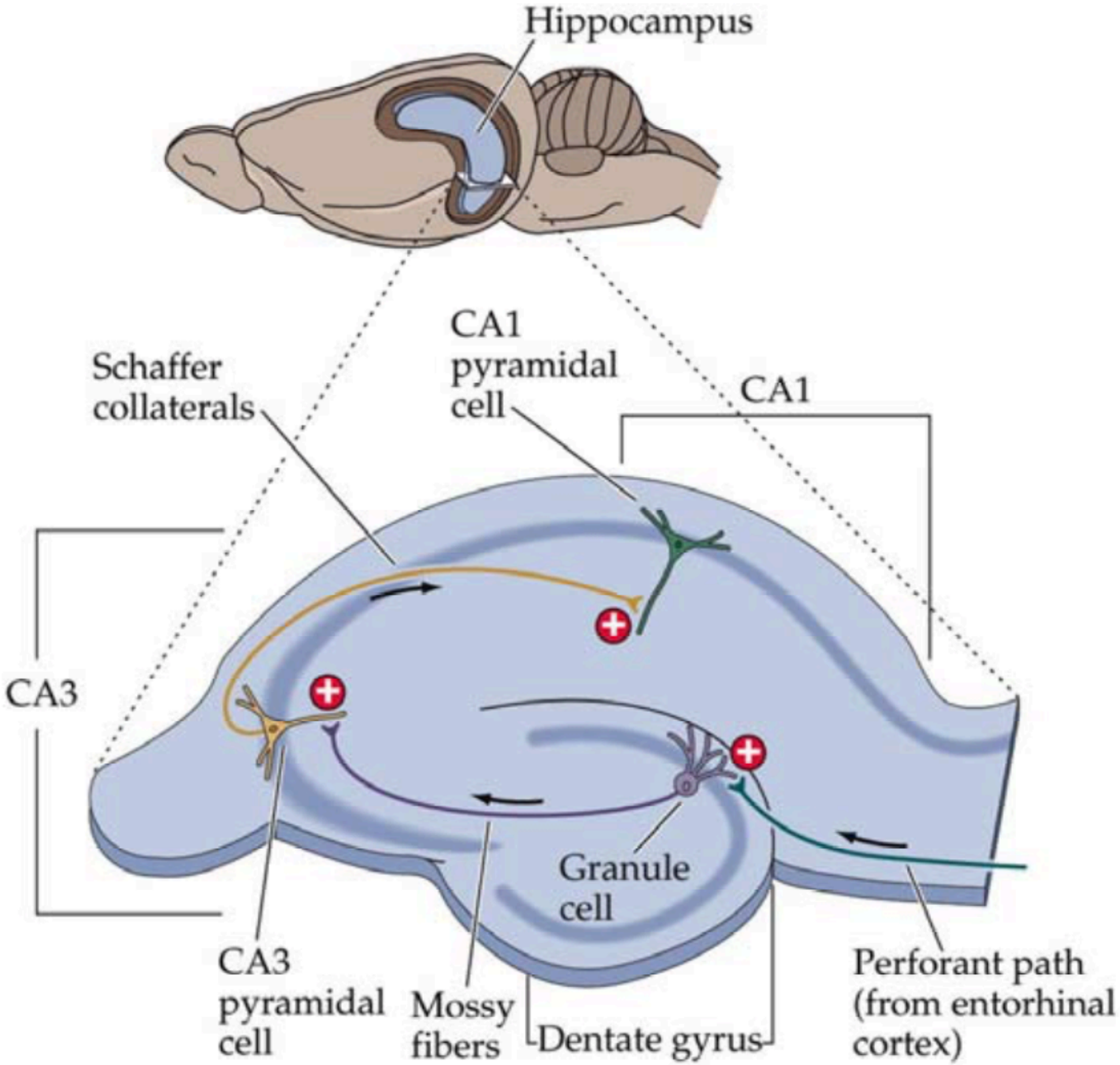
NMDA receptors in HC blocked



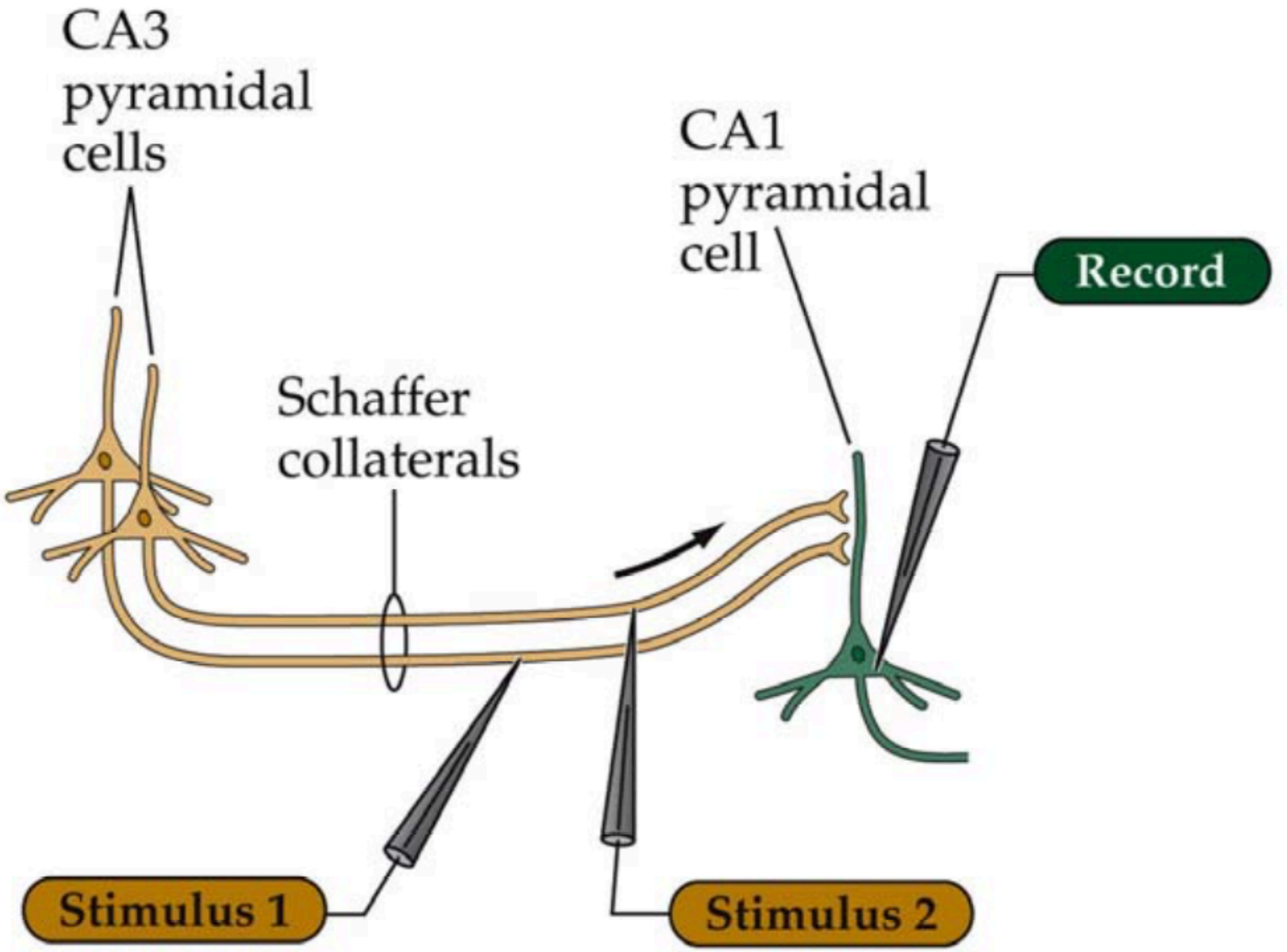




Most Studied Syanpse in Hippocampus: CA3>CA1.

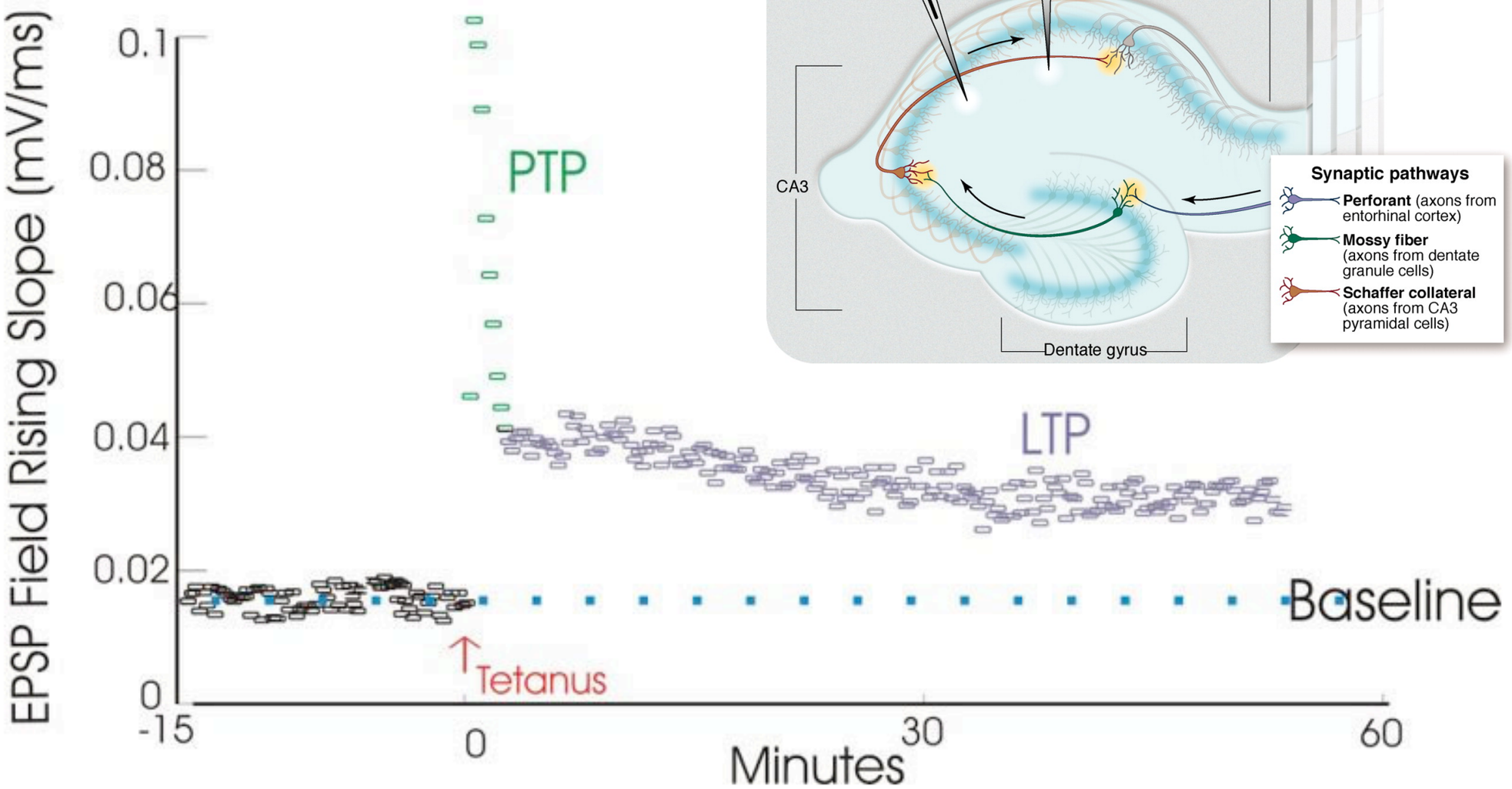


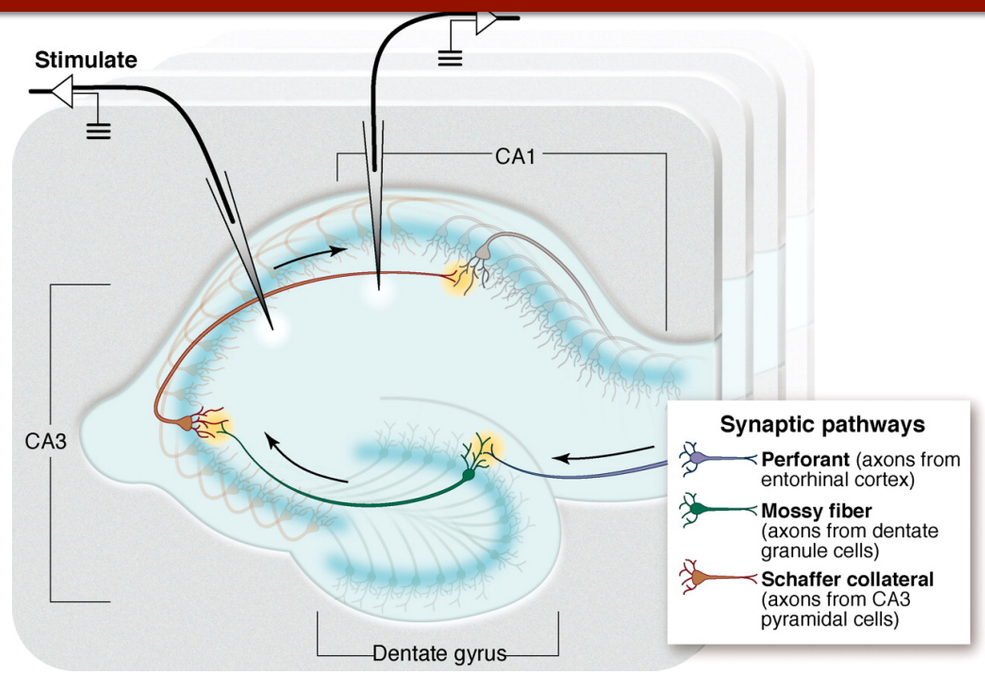
Most Studied Syanpse in Hippocampus: CA3>CA1.



Squire 2013, p1017

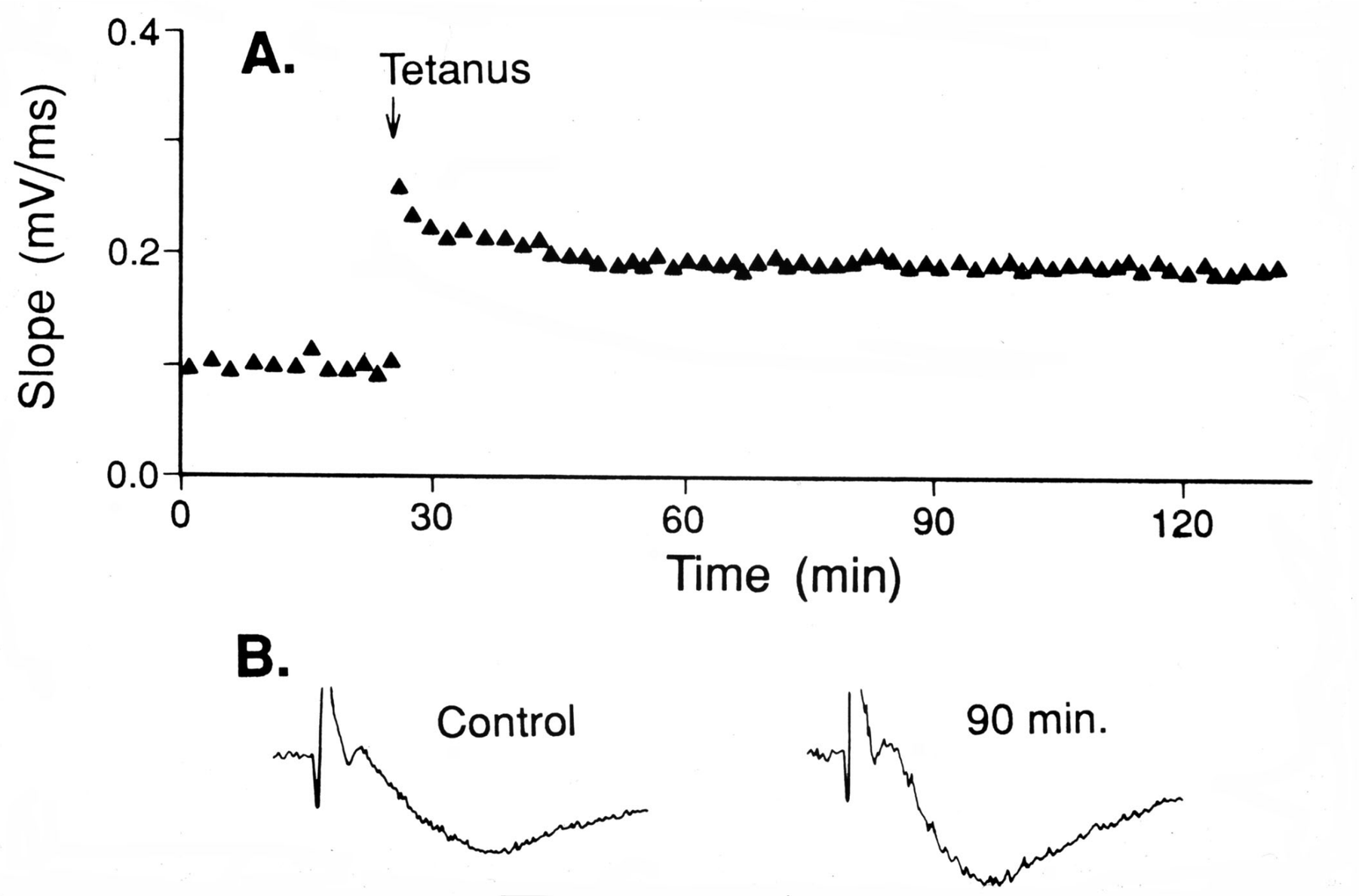
Post-Tetanic and Long-term Potentiation (LTP).





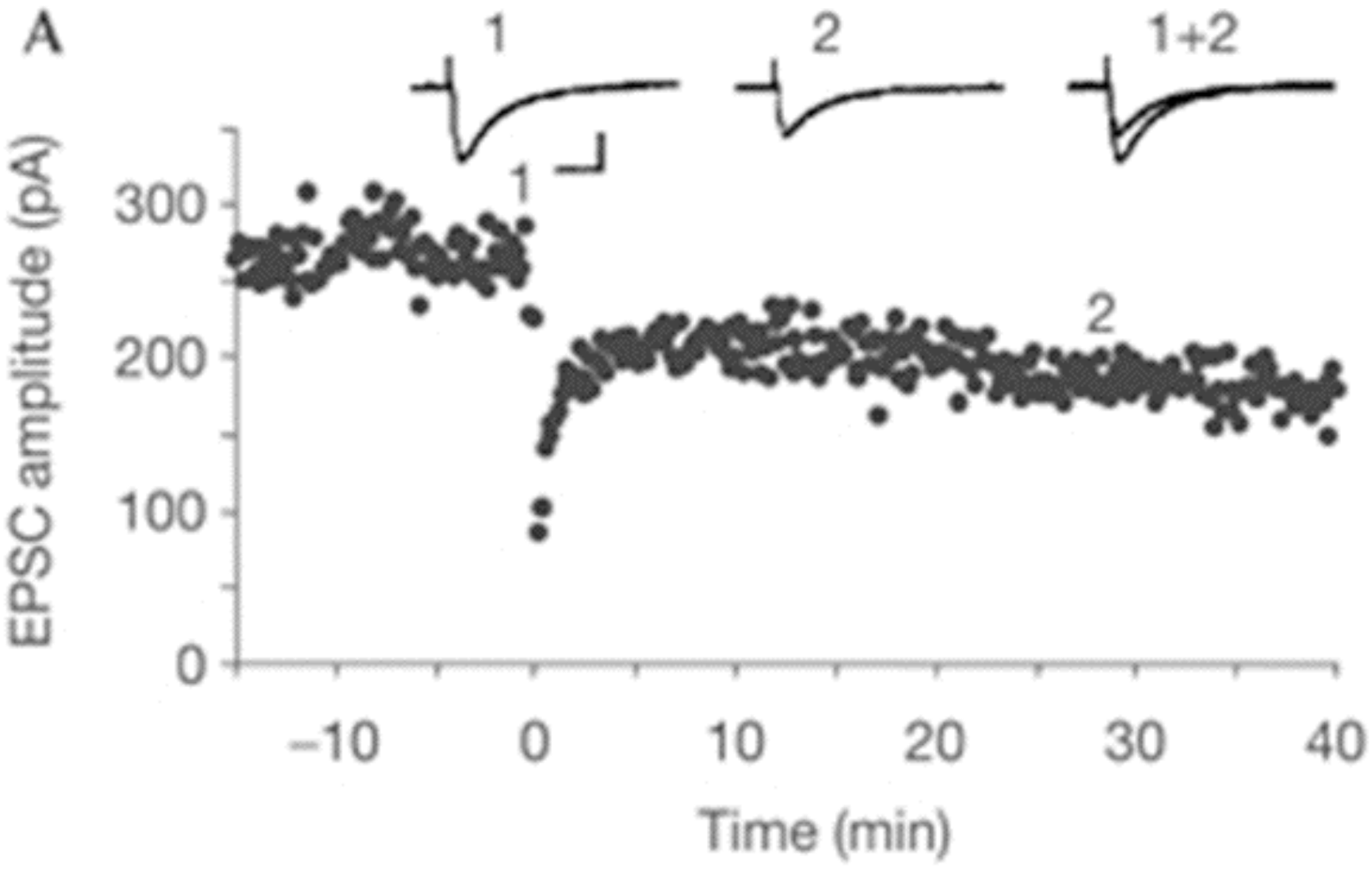
PTP believed to be caused by a large accumulation of Ca^{2+} in the terminal caused by a high frequency tetanic stimulation.

In vitro recordings of Hippocampal LTP.



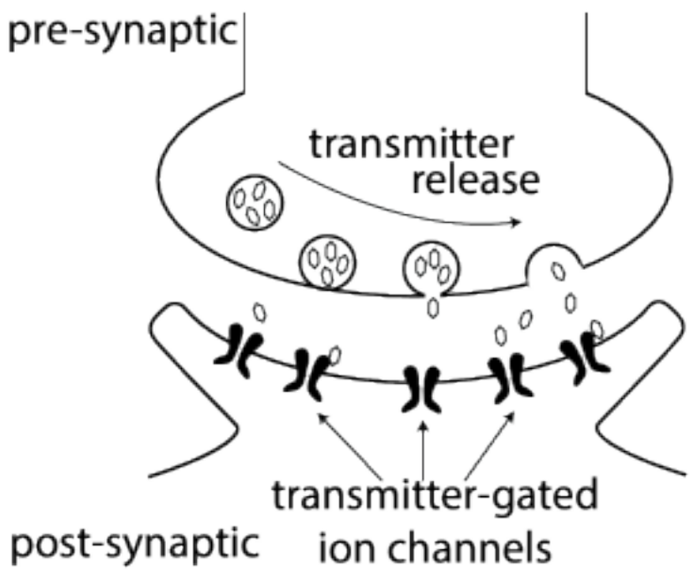
Stimulation frequencies that produce LTP usually range from ~50 to 200 Hz.

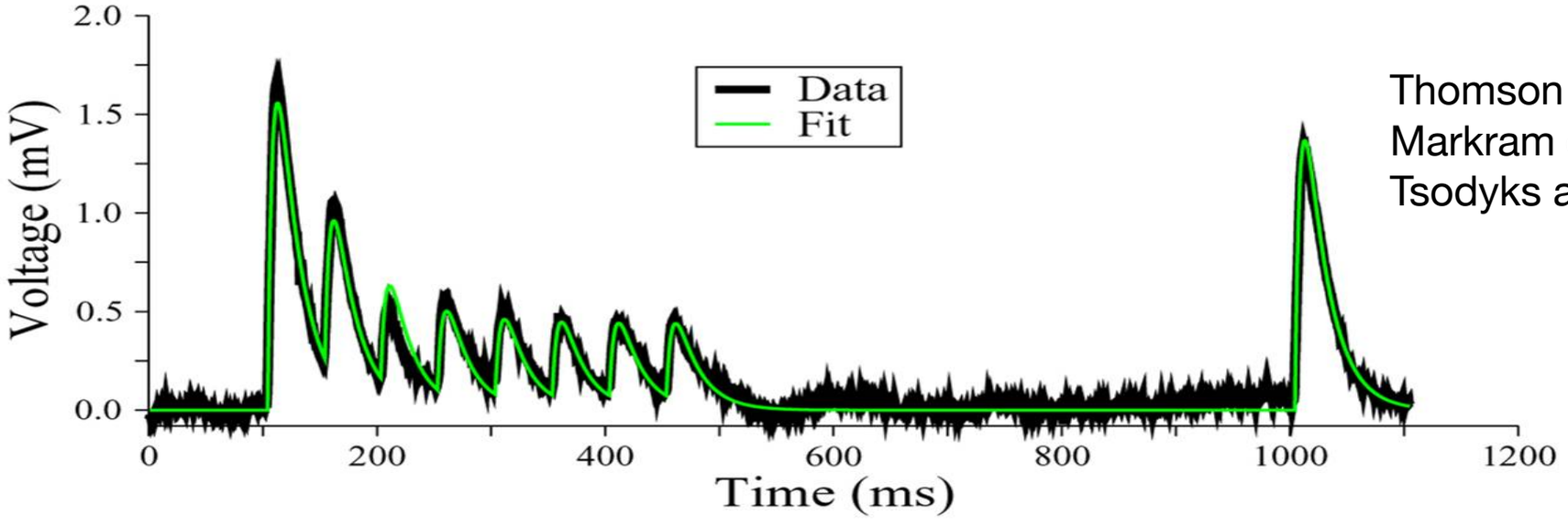
In vitro recordings of Hippocampal LTD.



Stimulation frequencies usually range from 1 to 10 Hz.

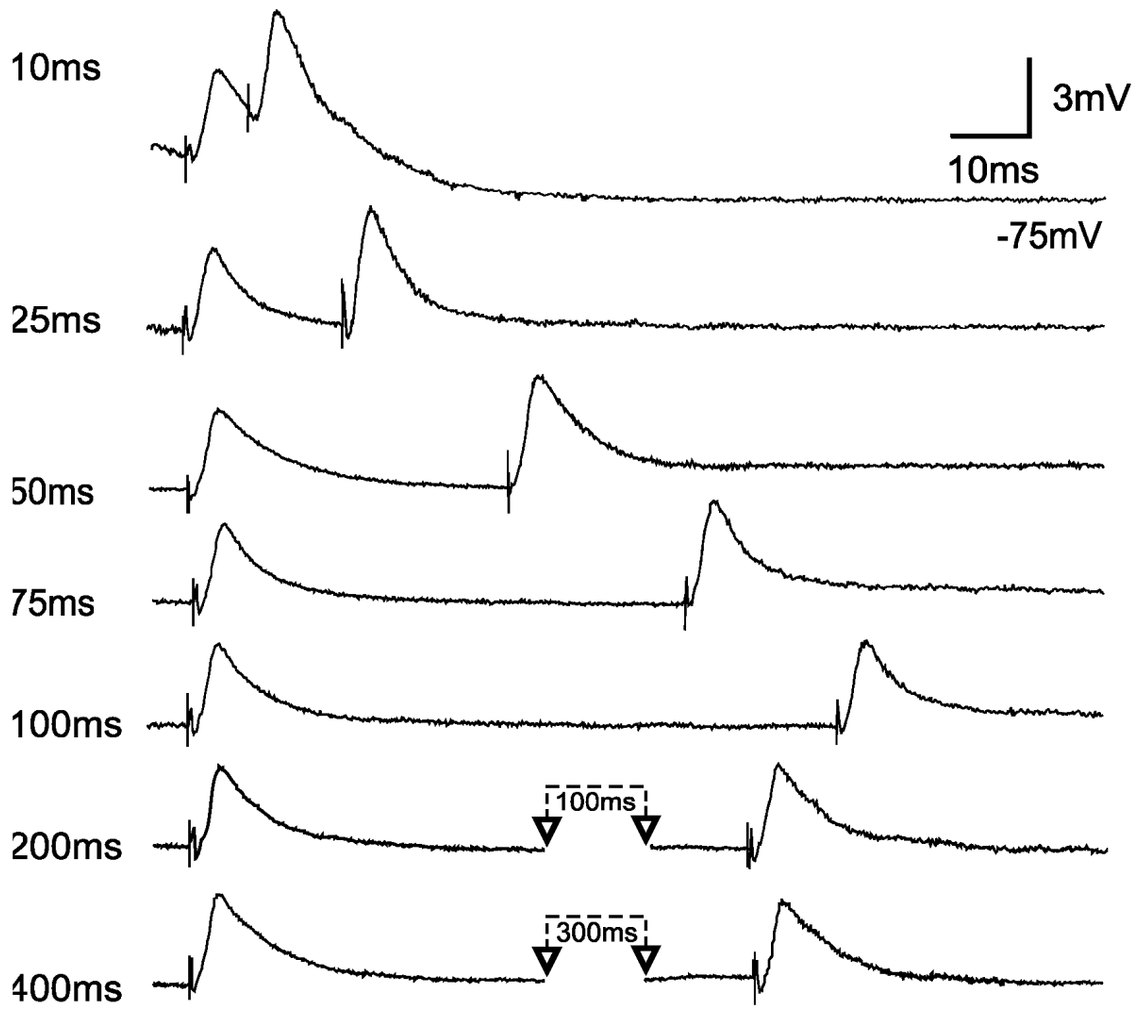
Short-Term Synaptic Facilitation/Depression.





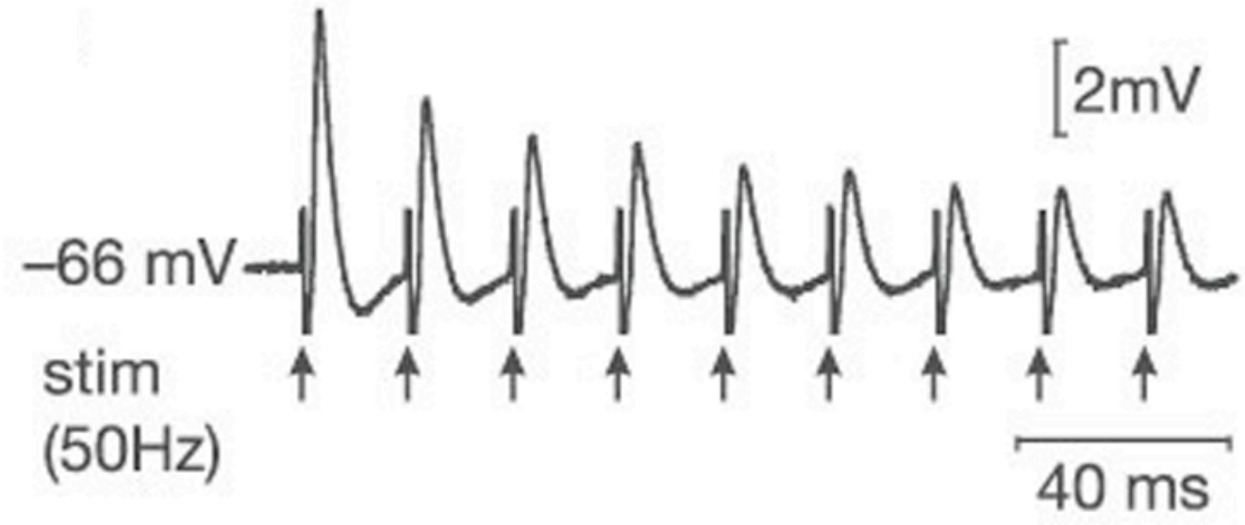
Thomson et al., 1993
Markram et al., 1998
Tsodyks and Markram 1997

Short Term Synaptic Plasticity: Paired Pulse Facilitation.

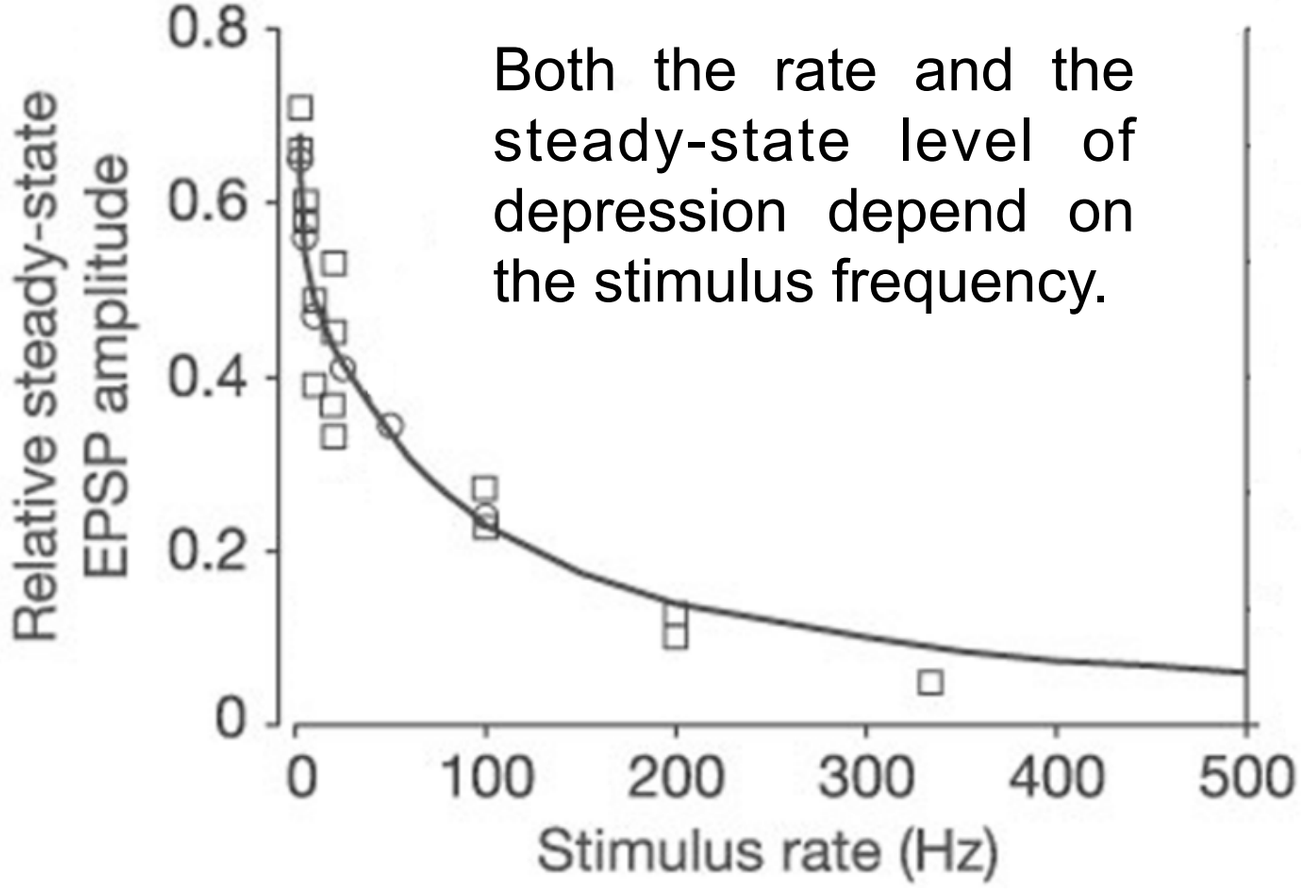


Paired activations of a synapse onto a CA 1 neuron. "Residual Ca^{2+}" in terminal for 10 to 100 ms after first stimulus increases probability of release.

Short Term Synaptic Plasticity: Synaptic Depression.



Successive stimuli at 50 Hz



Both the rate and the steady-state level of depression depend on the stimulus frequency.

Letter | Published: 02 January 2003

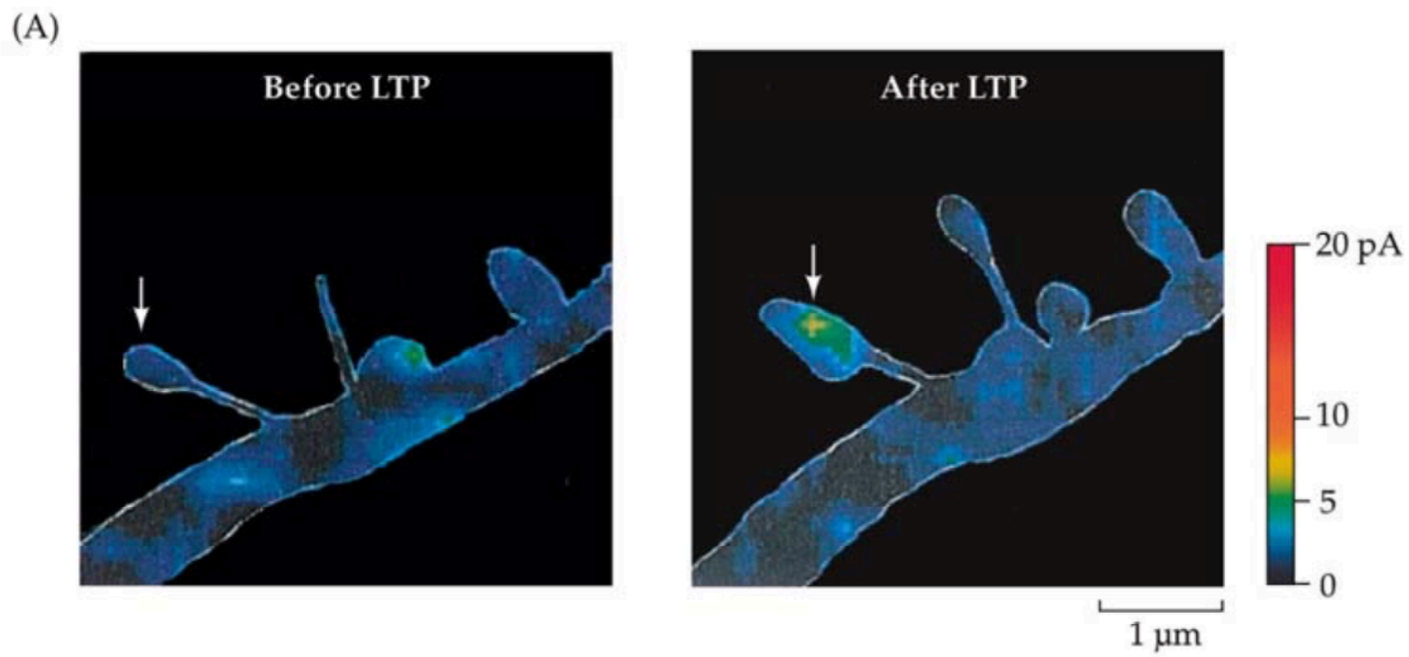
Synaptic depression in the localization of sound

Daniel L. Cook, Peter C. Schwindt, Lucinda A. Grande & William J. Spain

Nature **421**, 66–70 (2003) | Download Citation

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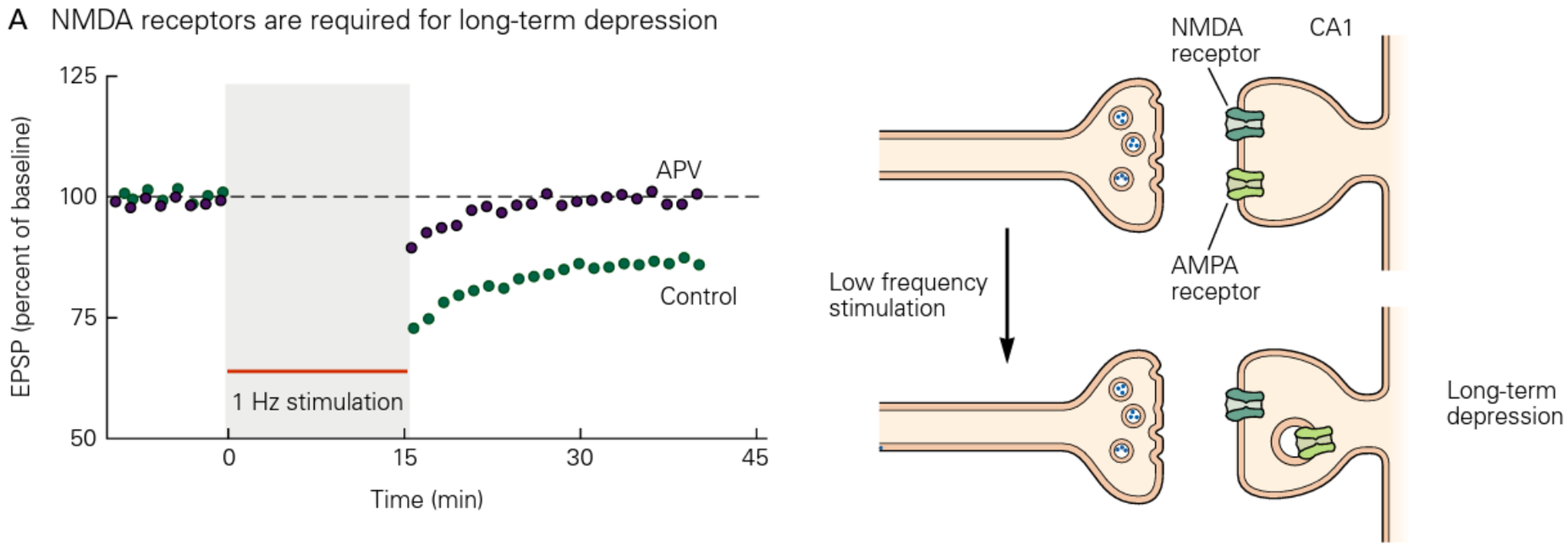
The Molecular Mechanisms Behind LTP and LTD.

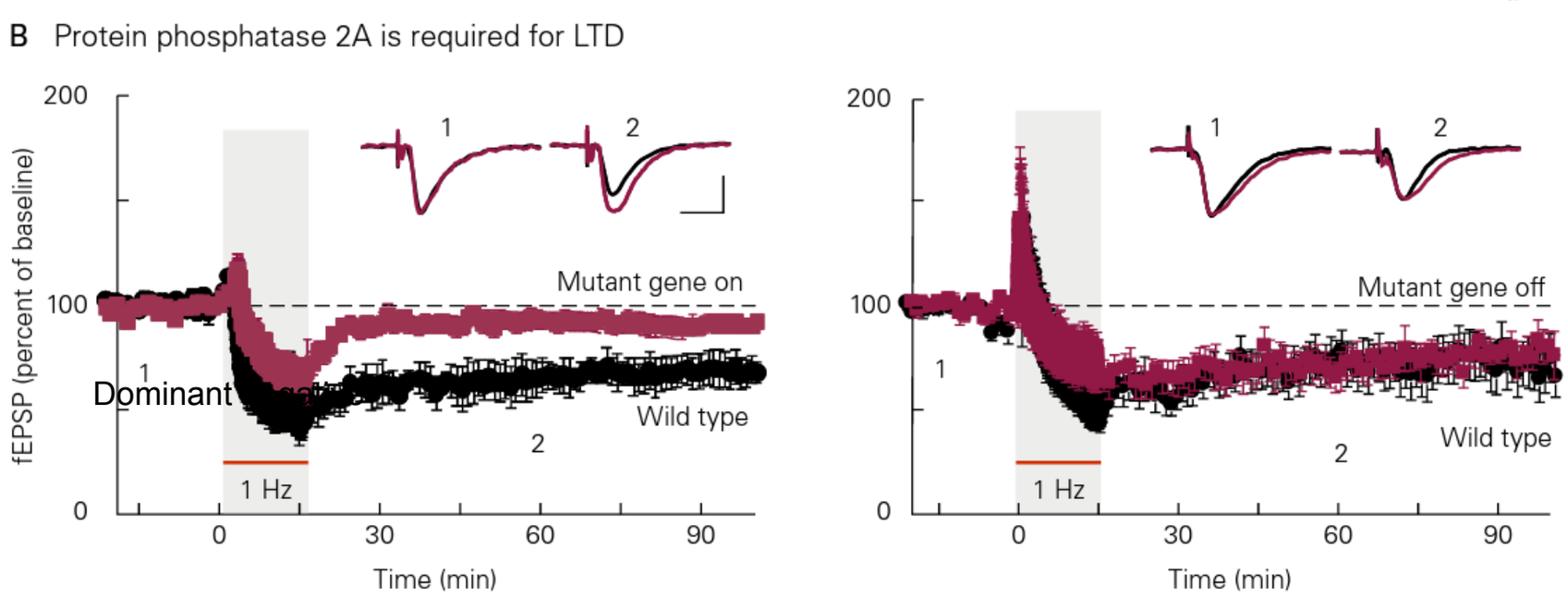


1. **Insertion of AMPA receptors into the postsynaptic membrane (LTP); or their removal from the postsynaptic membrane (LTD).**

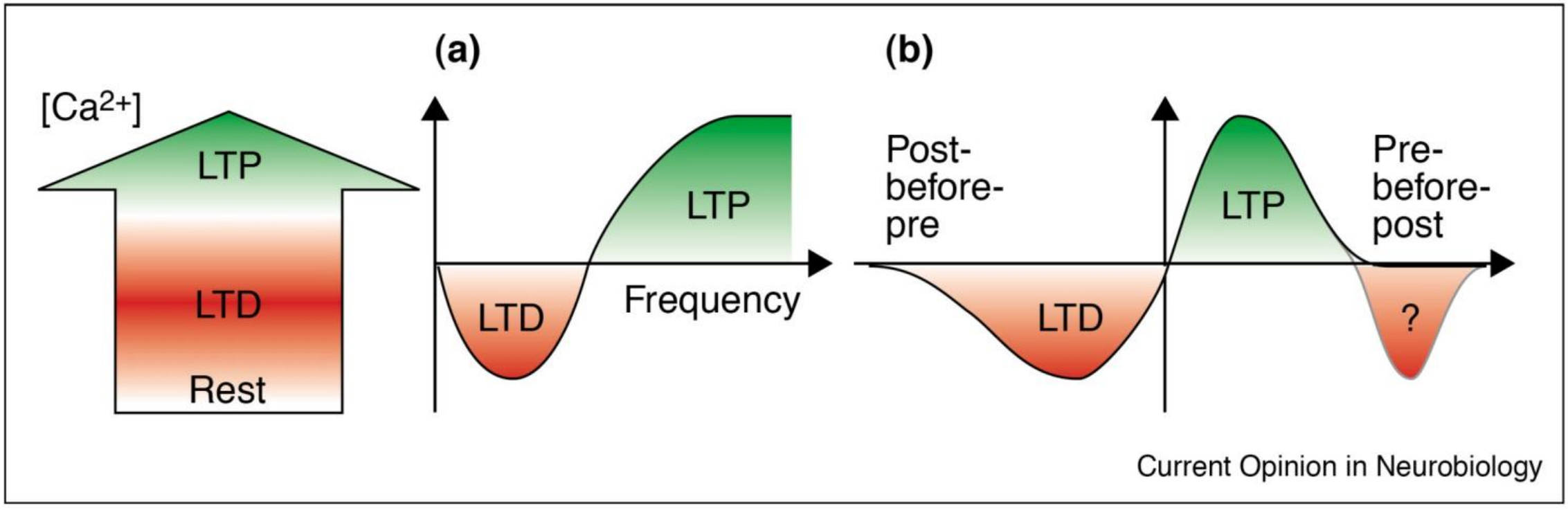
2. **Growth or shrinkage of the spine via reshaping of the actin cytoskeleton.**

The Molecular Mechanisms Behind LTP and LTD.





Synaptic Plasticity Strongly Depends on Calcium Levels.

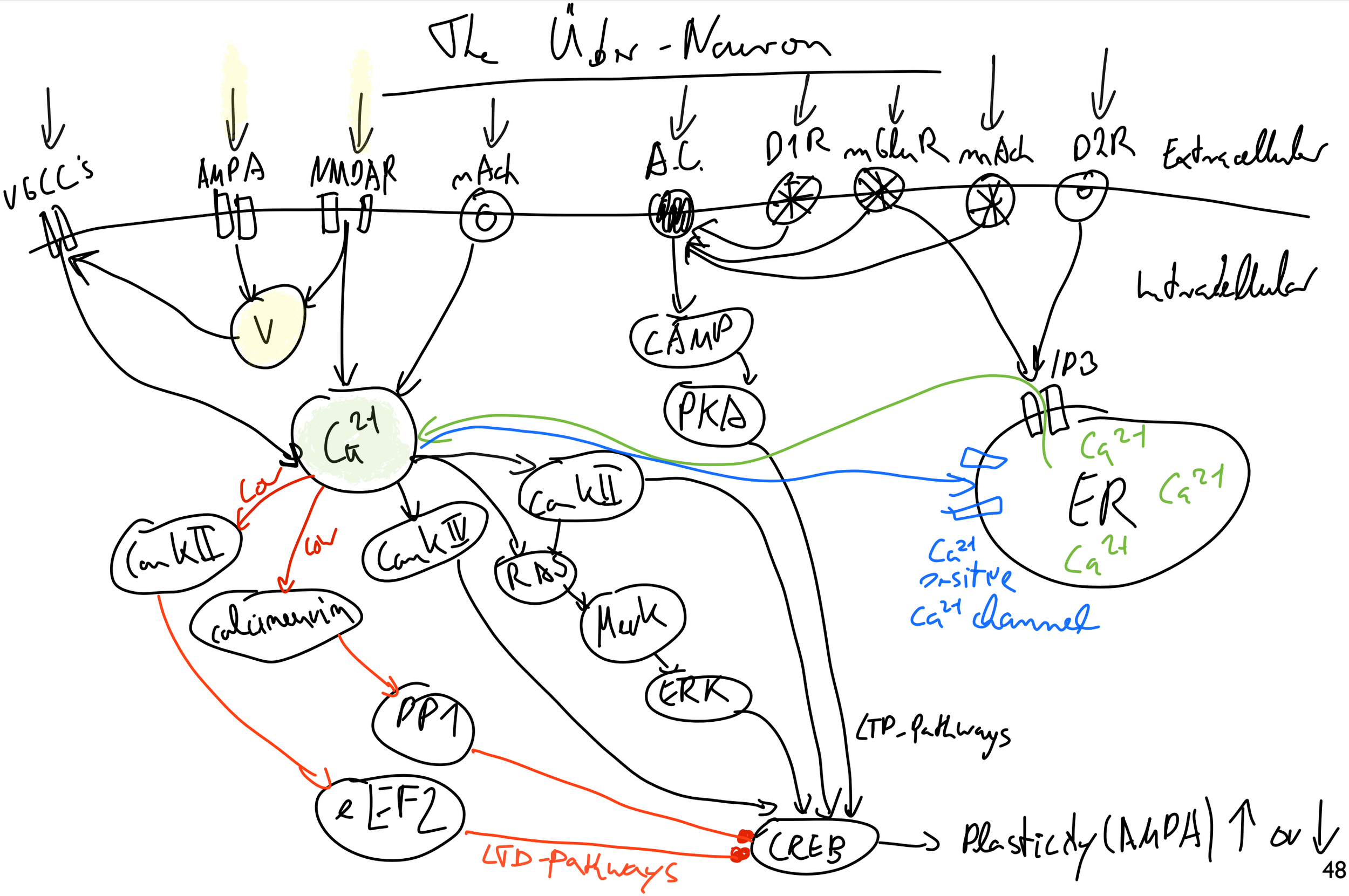


1. Level and timing of Ca^{2+} rise in spine determines LTD or LTP.

2. Low frequency synaptic firing (~5 Hz) produces LTD;
high frequency synaptic firing (~50 to 100 Hz) produces LTP.

3. The same Ca^{2+} rules may underlie
"spike-timing-dependent synaptic plasticity (STDP).

Intracellular Plasticity Signaling Pathways.



Thank you for your Attention!

